

INTRODUCTION

Progress is being made in creating a digital connection between service providers and customers through the use of technology as well as now delivering services digitally via mobile phones or the internet. Smartphone usage by individuals & the growth of affordable data have also changed how customers access services on an everyday basis. But despite this technological advancement, many of the local providers such as cab drivers, auto drivers, electricians, plumbers, carpenters & other daily wage workers still have difficulty accessing customers due to various factors.

In a number of local communities, particularly in semi-urban/rural locations, local service providers typically depend upon traditional means for connecting with potential customers, e.g., through word-of-mouth, their personal networks, and their physical presence at local market locations. These methods, however, are typically inefficient and untrustworthy (e.g., little to no verification of services), and provide little ability to quantify or assess the results of any potential job prior to the job being started. As a result, the majority of qualified workers have an inconsistent access to jobs, resulting in the underemployment of skilled workers.

Customers also face many challenges finding qualified & skilled service providers, especially when in a hurry or stressed. Due in large part to the absence of an organized and trustworthy platform, it becomes extremely difficult for customers to verify the quality of service being provided, compare available options, and to make informed decisions. The disconnect between service providers and customers created by the lack of a structured and digital solution is evident and continues to be a pain point for both parties.

The Deal Theka app is a unique solution that connects customers with local service providers through a single marketplace. The purpose of this app is to provide a digital marketplace where service providers can create profiles, promote their business, and connect customers with the service providers. Customers will also be able to search, review, and schedule for their services based on their individual needs.

Providers using the Deal Theka app will be able to create an account that includes a profile that lists their services and service offerings', pricing, and the availability to provide their services' to potential customers. Once a customer has created their account, they can then provide a filter to search and filter through the list of services by category of service and/or geographic location, read the full description for each service provider, contact the service provider via email or in-app, or directly book the service provider through the use of the app. This approach eliminates the need to find service providers using informal, unreliable means.

Similar digital marketplace platforms have been successful in various parts of the world such as Urban Company (previously known as UrbanClap), JustDial, and OLX Services, but tend to primarily focus on major urban markets, thereby not providing services to small towns or communities. Deal Theka's goal is to create a marketplace that is user-friendly, easy to navigate, and focused on the local needs of its users' even when the users may have very limited or no technical literacy.

Due to a heavy concentration of the Indian population living in tier-2 and tier-3 cities, there is a strong need for digital services that are appropriate to each area's local culture and environment (including linguistically). As Digital India continues to build awareness of the Internet in these areas and more people gain access to affordable smartphones, you will find that many of them will want to use their smartphones as a way of obtaining goods and services needed for daily living. Deal Theka is committed to creating opportunities for everyone to obtain goods and services through our platform.

In the development of our application from a technological standpoint, we used Flutter as the framework for creating cross-platform mobile apps with one code-base. Flutter helps us build responsive (able to adapt to the type of device being used) and visually attractive user interfaces (UIs). This means that all users, regardless of what type of device you are using, enjoy similar experiences with the application due to the use of Flutter. We also utilize Firebase and cloud-service based technologies for handling data storage, user authentication, and real-time updates on our back-end platform.

The Deal Theka platform was built with the goal of providing a scalable and flexible solution to growing demand for online services. Dynamic service attributes, real-time booking updates, secure authentication, data management, and modular architecture are all key elements of this system that make it capable of supporting a large number of services and users.

The technological advantages of the Deal Theka platform are complemented by significant economic and social benefits. By providing local service providers with a digital platform to promote their services, the Deal Theka platform will increase employment and income stability for local residents. Through expanding their reach beyond their local markets, service providers will be able to reach a larger customer base. Likewise, by utilizing a single source to find services, customers will save time searching for services; receive higher-quality service; and enjoy more convenience when utilizing services.

The platform creates transparency and trust for all users of the Deal Theka service by enabling service users to see details of each service they are being provided, as well as ratings and reviews that help build customer loyalty. This promotes service providers to continue to operate with high-quality service levels, which will ultimately lead to satisfied customers. Additionally,

the Deal Theka platform assists in the digital transformation of the local economy by increasing the use of technology in everyday activities.

Another benefit of the Deal Theka system is its ability to generate reports and monitor user activity. The platform provides a complete set of reports that include user reports, service reports and booking reports for administrators to use to review system activity and establish metrics to measure the level of success of the service. This data can also be used to identify trends, enhance services provided and increase the overall efficiency of the system.

In summary, the objective of the Deal Theka platform is to bring together service providers and service users in an efficient, effective and centralised manner to create a high-quality, efficient and user-friendly platform to match the traditional service discovery. It has been developed to overcome the challenges associated with the traditional method of discovering services, through the use of modern forms of technology to create a scalable and impactful solution. Deal Theka's goal is to increase access to services, reduce inefficiencies and assist local workers, while supporting both local and national economic and digital development.

OBJECTIVE

Through the Deal Theka project, we aim to provide an application that connects individual consumers and local businesses with a more efficient, accessible and reliable means of connecting with each other through a mobile app. The need for easy ways to find and use local services is increasing daily as the pace of digital change continues to accelerate. Conversely, many local businesses rely upon traditional methods of attracting customers and gaining new clients, which can limit their outreach and potential for growing service offerings.

The purpose of the Deal Theka system is to connect service providers and consumers through an integrated digital marketplace, thereby enabling service providers to showcase their products and provide services that the consumer is searching for. The application will be built with easy access in mind and will offer greater efficiencies to service providers and will also create an enjoyable experience for both providers and consumers when utilising the application, which will anecdotally reduce the likelihood of negative consumer experiences.

1. Create A Digital Platform For All Local Service Providers

One key goal of this project is to create a digital platform where all local service providers can register on and display their services online. These providers are workers such as taxi drivers, auto rickshaw drivers, electricians, plumbers, carpenters, mechanics, and any other workers who are paid hourly. The way this platform will work for these service providers is that you will be able to:

- Create Your Own Professional Profile
- List All Your Unique Services Description & Pricing
- Reach More People Beyond Your Current Network

By having the ability to create a digital presence for themselves, these service providers will have access to larger amounts of visibility and opportunities as opposed to being limited to the physical world.

2. Streamline The Process of Finding Quality Service Providers

The second key goal is to make it easier for users of the platform to find reliable service providers. In the traditional way, customers have always had to rely on word-of-mouth referrals or spend a significant amount of time searching for service providers manually. This can be both time-consuming and inefficient. The Deal Theka source provides a good example of how this process will be streamlined by:

- Giving users access to a searchable service provider database
- Allowing users of service provider database to filter service provider results based on individual categories & locations
- Displaying relevant data/information related to each individual service provider

Through the use of the above examples, users of the Deal Theka platform will be able to easily find quality service providers with minimal time or effort.

3. Create a Unified Service Directory System

The project will integrate existing service platforms into a single platform for accessing all services from one source rather than having to find services from multiple sources. Benefits of this centralized service directory system include the following:

- Decreased reliance on using multiple services/platforms to access services
- Improved access to services
- Structured and organized service directory database

By integrating services onto one platform, the Deal Theka application increases user convenience and ease of use.

4. Create Jobs for Local Workers

An essential part of the Deal Theka project includes providing opportunities for local workers to obtain employment. Many skilled workers are having difficulty securing reliable employment as they have limited exposure to potential employers. The Deal Theka platform is serving to provide:

- A digital presence for service providers
- Increased job opportunities
- Ability to generate more income

These factors will contribute to the overall growth of the local economy and the ability to sustain themselves as productive members of the community.

5. To reduce the time and effort in hiring service providers

The application has been developed to help minimize the amount of time and effort that will be needed to hire a service provider or service; therefore, purchasers can search, request quotes from and book service providers all within the same application. Some advantages of this include:

- Fast discovery of service providers
- Instantaneous booking of service providers
- Minimal reliance on manual processes

This ultimately increases the efficiency of both the customer and the service provider which increases customer satisfaction.

6. To provide an easy-to-use mobile app

The app has been developed with a focus on usability. Most of the users of this application will not have extensive technical knowledge; therefore the app's design allows for a simple and intuitive user interface. Features of this app include:

- Desktop-like interface design
- Easy-to-use navigation
- Simple, clear instructions and input screens

These features will make it possible for many different types of users (including users who have little or no technical knowledge) to use the application.

7. To ensure the system is secure and reliable

One of the key objectives of the system development is to provide a secure and reliable application. The application will provide protection of user data and will provide users with consistent operation of the application. Security controls include:

- User authentication
- Data verification
- Control of user access to application functionality

Providing these controls will help build trust between users and will enhance the user's perception of the reliability of the application.

8. Improving operational efficiency in managing service requests

The Deal Theka system was designed to provide the service provider and administrator with effective tools for managing their services and the bookings associated with those services through the use of the following functionality:

The administration can manage services through:

- Adding new services
- Updating existing services
- Managing requests for bookings of specific services
- Tracking the status of services

This functionality creates greater operational efficiency within the administration and provides more control of the system.

9. Supporting data-driven decision making for system performance

The Deal Theka system includes several reporting capabilities that allow the administration to evaluate the system's performance. Reporting enables the administration to identify:

- New users joining the system
- Services being used
- Trends in booking of services

The data generated through these report capabilities can be utilized to improve system performance.

10. Delivering a scalable and future-ready platform

A primary goal of designing the Deal Theka system was to provide the foundation for future scalability. The architecture of the Deal Theka system utilizes modern technology that will support its growth and expansion. The following are some examples of future enhancement possibilities:

- Online payment processing
- GPS-based service providers within the region
- Machine Learning-Based suggestions to improve revenue production

The design of the Deal Theka system ensures that it can evolve with the demands of the future.

Deal Theka's project vision is to create a dependable, efficient, and easy-to-use digital system to connect service providers with customers. The project uses modern technology to increase access to services, reduce waste with the existing system, and to create new jobs for local people.

The project creates efficiency through the use of a new platform to enable users to connect to service providers and be part of their digital transformation. The platform will be a sustainable model that creates a mutually beneficial relationship between service providers and users and helps to create a more connected and efficient environment.

System Analysis

System analysis is the process where the current situation is studied, problems are identified and the requirements of the new system are established. It assists in knowing the needs of the user, the shortcomings of the system and its viability before real development takes off. In the case of the **Deal Theka** project, system analysis was done to determine the gap existing in the current service discovery process, as well as to suggest an effective digital approach.

1. Identification of Need

Identification of need is based on the need to know why the proposed system is necessary and what gaps are present in the current environment.

Getting credible service providers in most local communities like electricians, plumbers, drivers, carpenters and other daily workers is usually a difficult undertaking. Informal searches like personal references, neighborhood contacts or manual searching are normally relied on by customers. These methods are not only time consuming but are not reliable and transparent whereby the user cannot easily check on the quality or availability of services.

Conversely, a good number of talented employees and service providers struggle to tap more customers. They can only have local exposure and sporadic job opportunities due to the lack of an organized digital platform. This limits their growth and income stability opportunities.

As the number of people using smartphones and having access to the internet, particularly in the semi-urban and rural regions, there is an increasing need to have digital solutions that can help streamline daily activities. It is possible to have a centralized mechanism that links customers to the service providers and this can greatly enhance efficiency, save time and make the use more convenient.

Thus, there is a necessity to fill the gap between the customers and service providers by providing a credible, convenient and easy to use digital marketplace of local services, which can be represented by such a platform as **Deal Theka**.

2. Preliminary Investigation

It is done through preliminary investigation that is done to check the feasibility and practicality of the proposed system. It is the process of understanding the existing system, determining

limitations of the existing system and determining whether the new system can adequately solve the problems.

2.1 Study of Existing System

The current system on locating local services is highly manual and unstructured. Customers rely on:

- Word-of-mouth recommendations
- Personal contacts
- Local advertising or going to the shopping malls.

Likewise, service providers rely on:

- Local visibility
- Limited customer networks
- Informal communication

2.2 Weaknesses of Current System

The existing strategy has a number of weaknesses:

- **Laborious process:** It is laborious to seek appropriate service providers.
- **Deficiency of reliability:** No adequate checking system or rating.
- **Poor reach:** The service providers have a poor reach to a larger audience.
- **Lack of a centralized platform:** The information is distributed and not organized.
- **Lack of efficient communication:** No user-provider efficiency on communication.

2.3 Feasibility Study

The proposed Deal Theka system was evaluated based on different feasibility aspects:

a) Technical Feasibility

Modern technologies like web/mobile frameworks and relational databases can be used to develop the system. The tools and technologies needed are easily accessible and can be used.

b) Economic Feasibility

The project is cost-effective since it can be built with open-source software and a small amount of infrastructure. Maintenance expenses are also affordable.

c) Operational Feasibility

The system is to be customer-friendly and easily accessible to customers and service providers with the minimal technical skills. This makes it easily adopted.

d) Social Feasibility

The system is advantageous to both users and service providers in that it enhances access, augmented employment opportunities, and wastage of time. It is consistent with the increasing trend of online platforms in the local societies.

2.4 Proposed System Overview

The **Deal Theka** system is an online platform which enables:

- Service providers to add and post their services.
- Search and get services available to the customers.
- Users to see the details and choose the services according to their needs.

The purpose of this system is to develop a well-organized and effective atmosphere in which both sides will be able to communicate effectively.

The system analysis clearly indicates the need for a digital solution to overcome the limitations of the existing manual system. The initial research proves that the suggested **Deal Theka** platform is both technically possible, economically viable, and operationally feasible. It can greatly enhance the manner in which local services are accessed and handled.

3. Feasibility Study

3.1 Technical Feasibility

The **Deal Theka** mobile app is built on Flutter that is popular in creating modern cross-platform mobile apps. Flutter enables developers to build both Android and other applications with a single codebase and makes the process of development efficient and scalable.

The technologies needed in this project are:

- **Flutter** mobile application development.
- **Dart programming language** to application logic.

Backend server services and application program interfaces (APIs) to handle user and service provider data.

- User profile, service listing and booking information database systems.

These are a wide range of technologies, thoroughly documented, and have a big developer base. Moreover, the developer is also familiar with mobile application development and programming, which also contributes to the effective realization of the system.

Thus, technically, the creation of the **Deal Theka** platform is possible and attainable with the help of the tools and technologies available.

3.2 Economic Feasibility

Deal Theka project does not need a lot of financial investment as it is mainly founded on software development. The technologies utilized in the project, including Flutter, Dart and open-source development tools are all free. This goes a long way in lowering the cost of development.

The key costs related to the project can be:

- Internet connectivity
- Google Play Console Account
- Cloud Storage Cost
- Mobile device testing

The overall cost of developing the project is low because the project is done as an academic industrial project and the technologies used are open-source. Consequently, the project would be cost efficient.

3.3 Operational Feasibility

The **Deal Theka** application will be easy to use and understand to ensure that service providers and customers can use the platform without the need to have advanced technical expertise. Service providers find it easy to register and post their services and customers can search and book services using a simple interface.

Common services that the application is supposed to assist in include transportation, home maintenance, repair services, and daily labour. As smartphones are the most common devices nowadays, users can comfortably access the application at any time and place.

The system also enhances customer convenience as customers can easily locate close-by service providers. Meanwhile, it assists local workers to be more visible and reach potential customers.

The **Deal Theka** system is operationally feasible due to its simplicity, accessibility, and usefulness in practice.

3.4 Schedule Feasibility

The **Deal Theka** project is being created as a final year industrial project and the development of the project is scheduled to be based on the academic calendar. The process of developing the project consists of many steps like requirement analysis, system design, application development, testing, and documentation.

As the project is developed using the current development models such as Flutter and modular design approach, the development time can be achieved with ease and within the stipulated time.

This is achieved by proper planning, testing on a regular basis and systematic development of the project so that the project is completed successfully before the end date of submission.

4. Project Planning

Project planning is a vital step that maps out the development process of the **Deal Theka** application, its management, and its finalization within a specific time frame. It involves scheduling, resource allocation, methodology of development and allocation of tasks. Planning properly will make the project efficient, time-bound, and it will achieve all the objectives that are set.

The **Deal Theka** mobile application will be developed in a 12-week development cycle in a systematic process using the Software Development Life Cycle (SDLC).

4.1 Development Model

The project is based on the Waterfall Model, according to which the development will be carried out in a number of clearly defined stages.

Why Waterfall Model?

- Appropriate in scholarly projects having predetermined needs.
- Unproblematic to maintain and record.
- Every stage is accomplished prior to the next one.

4.2 Project Phase

The following key phases will be used as the development process:

Phase No.	Phase Name	Description
1	Requirement Analysis	Determine the issues encountered by the customers and the local service providers and establish system requirements of the Deal Theka platform.
2	System Design	Design the system architecture, database, UI flow and application modules within the mobile platform.
3	Implementation/ Development	Build the mobile application in Flutter and Dart, add Firebase services, and add the basic modules (user registration, listing of services and booking).
4	Testing	Unit testing, integration testing and user testing should be conducted to make sure that the application is working properly and efficiently.
5	Deployment	Install the mobile application to test and demonstrate it in Android devices.
6	Documentation	Write the final project report, user documentation, and presentation materials and submit them.

The planning of the project of Deal Theka makes sure that there is a systematized approach to development. The project can be completed within the stipulated time and retain quality and functionality by breaking the project down into phases, allocating resources effectively, and handling risks.

5. Project Scheduling (12 Weeks timeline)

Project scheduling involves organizing activities, determining the order of their completion, and estimating time to be taken to complete them. It assists in tracking the progress, schedule control and so that the project does not take longer than it was scheduled to take. In case of the Deal Theka mobile application, two commonly used methods of project scheduling are used:

- **PERT (Program Evaluation and Review Technique)**
- **Gantt Chart**

These tools give a graphical as well as an analytical perspective of the project schedule and dependency of its tasks.

5.1 PERT Chart (Program Evaluation and Review Technique)

PERT is a project management tool that is employed in examining and illustrating the order of activities undertaken in the completion of a project. It is concerned with finding dependencies among activities and the time that is minimally necessary to complete the project.

The second activity is the estimation of time required to perform the activities.

The key activities of the **Deal Theka** project and the approximate time of these activities are as follows:

Activity ID	Activity Name	Duration (Weeks)	Predecessor
A	Requirement Analysis	2	—
B	System Design	2	A
C	Implementation	4	B
D	Testing	2	C
E	Deployment	1	D
F	Documentation	1	E



5.2 Gantt Chart

A Gantt chart is a pictorial representation of the project schedule; the tasks as compared to time. It assists in monitoring the progress and learning how work overlaps.

5.2.1 Task Scheduling Table

Project completion activities will be involved as shown on a weekly basis below:

Week	Activity	Description
Week 1-2	Requirement Gathering & Analysis	Develop a user requirement, project scope, and system functionality like service listing, search, and bookings.
Week 3-4	Design Phase	Design system architecture, create ER diagrams, DFDs and UI layouts of the mobile application.
Week 5-8	Development Phase	Phase Develop application modules like user registration, service provider profile, service search, booking system and admin management.
Week 9-10	Testing Phase	Do functional testing, integration testing and bug fixes to enhance system performance.
Week 11	Deployment	Deploy the application to demonstrate and test it on Android devices of the final test.
Week 12	Documentation & Presentation	Create final project documentation, presentation slides and demonstration materials.

5.3.2 Gantt Chart (Timeline Representation)

Activity	Week 1-2	Week 3-4	Week 5-8	Week 9-10	Week 11	Week 12
Requirement Analysis						
System Design						
Development						
Testing						
Deployment						
Documentation						

Legend:

 = Task Duration

Software Requirement Specification (SRS)

1.1 Purpose

This document aims to give an elaborate outline of the requirements of the Deal Theka application. It establishes non-functional and functional requirements, system features and system constraints. This document will be used as a guide by the developers, testers and stakeholders in knowing how the system behaves and what to expect.

1.2 Scope

Deal Theka is an online application that will be installed on mobile devices that will allow customers to get in touch with local service providers like electricians, plumbers, carpenters, drivers, and other day-to-day workers. The system offers a centralized point at which service providers can create and advertise their services, and customers can search and see the services and choose them according to their needs.

The application will seek to:

- Make it easier to locate local services.
- Provide visibility to service providers.
- Facilitate effective interaction among users and providers.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirement Specification
UI	User Interface
API	Application Programming Interface
DBMS	Database Management System
SDLC	Software Development Life Cycle

This document outlines the general system functionality, system features, user requirement, and performance expectations of the Deal Theka application.

2. Overall Description

2.1 Product Perspective

Deal Theka is an independent application that can be characterized as a digital platform between two categories of users:

- Customers
- Service Providers

It communicates with a back-end database and can be configured with cloud services (like Firebase) to validate users and store their information.

2.2 Product Functions

The primary activities of the system are:

- Registration and login of users.
- Service provider registration

Service listing and service management.

- Search and filtering of services
- Viewing service details
- Reservation or call to service providers.

2.3 User Classes and Characteristics

User Type	Description
Customer	Browses and chooses services
Service Provider	Enroll and catalogs services
Admin	Handles system information and people

2.4 Operating Environment

- Android-based mobile devices
- Internet connectivity required
- Firebase or such like backend support

2.5 Design Constraints

- Less time of development (academic project)
- Use of open-source tools

- Compatibility with Android devices
- Internet dependency

2.6 Assumptions and Dependencies

- Basic smartphone knowledge of the users
- There is access to the internet
- Firebase or back-end services are not shut down

3. Functional Requirements

Functional requirements are what the system is supposed to do.

3.1 User Registration

- The system will enable users to make an account
- The system will authenticate user input (email, password, etc.)

3.2 User Login

- The system will have the ability to authenticate users based on valid credentials
- The system will not allow invalid login and show an error message

3.3 Service Provider Registration

- The system will enable service providers to enroll
- Service details (category, price, and description) shall be stored in the system

3.4 Service Listing

- The system will show a list of the available services
- The system will provide a filtering by category and price

3.5 Service Details

- The system will show elaborated data on the services of choice
- The system will indicate provider information

3.6 Booking/Contact

- The system will enable users to call or make reservations
- The system will have booking information

3.7 Admin Functions

- The system will enable the administration of users and services
- The system will enable records to be deleted or edited

4. Non-Functional Requirements

4.1 Performance Requirements

- Response time of the system is to be less than 2-3 seconds.
- Effective management of numerous users.

4.2 Security Requirements

- It is essential that passwords are encrypted.
- There should be limited access without permission.
- Input validation should be resistant to attacks.

4.3 Usability Requirements

- Simple and intuitive user interface.
- Easy navigation
- Minimal learning required

4.4 Reliability Requirements

- System should function without crashes
- Information is to be stored properly.

4.5 Scalability Requirements

- System must be able to accommodate expansion
- Has ability to support more users

5. External Interface Requirements

5.1 User Interface

- Mobile-friendly interface
- Clean and simple design
- Interactive screens

5.2 Hardware Interface

- Smartphone (Android)
- Internet connectivity

5.3 Software Interface

- Backend services (Firebase)
- Database management system

5.4 Communication Interface

- Internet-based communication
- Backend and frontend API communication

6. System Features

Feature 1: User Management

- Register, login, logout
- Profile management

Feature 2: Service Management

- Add, update, delete services
- Categorize services

Feature 3: Search and Filter

- keyword Search services
- sort by category or price

Feature 4: Booking System

- View service details
- Contact or book services

7. Future Enhancements

- Online payment integration
- Rating and review system
- GPS-based service suggestions
- Notification system

The Software Requirement Specification of the Deal Theka application is a detailed document outlining the functional and non-functional requirements of the system. It gives a precise insight on system behaviour, interactions between users and constraints. The document serves as a guide to system design, development and testing so that the end product is effectively developed to meet the needs of the users.

System Architecture

The architectural model of the Deal Theka system defines the global structure of the Deal Theka application and illustrates how its components interact with one another in order to achieve the deal theka application and deal management. At the architectural model level, there is an understanding of the complete functionality of the Deal theka system by providing the following high-level representation of the structure of the Deal Theka application:

The Deal Theka system is using a client-server architectural model combined with a cloud-based back-end infrastructure to enable the mobile application to act as a client and for the Firebase cloud-based services to act as the back-end and application services infrastructure for the Deal Theka application. This model provides for scalable, reliable, and real-time data processing and storage.

The main objectives of the architecture of the Deal Theka system are to deliver:

- Smooth user interaction;
- Safe handling of data;
- Fast processing of data and returning the results to the client;
- Future enhancement scalability.

Overall Architecture Overview:

The Deal Theka app architecture is built on a three-tiered model consisting of the following layers:

1. Presentation Layer – Frontend
2. Application Layer - Backend Services
3. Data Layer - Database

The flow from user to user is as follows:

User → Flutter mobile application → Firebase backend → database → response to the application → User

This layered model separates concerns for easier maintenance, upgrades, and scalability.

Architectural Diagram

The diagram shows the flow of interaction between users and the app, the app and Firebase services, and Firebase services, including authentication, database, and storage, and the flow of data back to the users.

System Architecture Components

Presentation Layer (Front-end)

The presentation layer is the user interface (UI) of the application. This will be built with Flutter and will be responsible for displaying information, as well as user interactions.

Some examples of features in this layer are:

1. User-friendly interface
2. Responsive Design for multiple device types
3. Screen navigation
4. User input handling (i.e. User Registration, Search, Bookings)

Some of the main screens in the presentation layer would include the following:

1. Login / Register Screen
2. Home Screen
3. Service Listing Screen
4. Booking Screen
5. Profile Screen

Using Flutter allows for the development of a cross-platform application from a single codebase and will provide a consistent experience across device types.

Application Layer (Back-end Services)

The back-end of the Deal Theka application is supported by the Firebase platform, which is a cloud-based service that supports application logic and data processing.

Below are a few of the key back-end services that will be used in the application:

1. Firebase Authentication

- Manages User Login / Registration.
- Provides secure authentication for users.
- Supports Email / Password Login.

2. Firebase Firestore (Database)

- Stores User Data, Service Listings, Booking Data.
- Provides real-time data sync for users.
- Enables quick access to user data.

3. Firebase Storage

- Used to store images (i.e., User Profile Pictures, Service Images).

Data Layer (Database)

The data layer is responsible for storing all the data for the application and managing the relationships between all the types of data. The data that will be stored includes:

- Users
- Service providers
- Available services
- Booking records

Database: Firebase Firestore will be used as the database.

Advantages:

- Live updates
- Able to support a large number of users
- Flexible structure of the database

System Architecture --- Workflow

Workflow describes how data moves through the system:

1. User Action

Users will be able to take the following actions within the mobile app:

- Create an account
- Search for services
- Book a service

2. Sending the Request

The app will send an API request to the Firebase services to retrieve data.

3. Authorization

Firebase will verify the user's credentials and authorize access to the application.

4. Retrieving Data

The data requested (such as available service(s)) will be retrieved from the Firestore Database.

5. Processing the Request

The request will be processed by the backend and an appropriate response will be generated.

6. Returning Data to Users

The application will return processed data to the mobile app for display to the user.

Benefits of the Model

1. Scalable

- Can accommodate multiple users
- Supports further growth potential

2. Reliable

- It has a high uptime rating due to its use of Firebase resources
- It provides a high level of data security.

3. Real-Time Data

- Users will receive instantaneous updates
- Users will have a great experience while using the application.

4. Cost-effective

- Software is free and/or open source
- Relatively little cost associated with hosting the system.

5. Manageable

- The modular design allows for easy expansion
- There are minimal obstacles to troubleshooting.

Security Issues Associated with the Architecture

It is extremely important to establish all of the necessary security precautions throughout the design of the architecture, as this is one area where users and the system will be protected from unauthorized access to their personal information. There are four types of security measures to consider when designing the architecture:

- User Authentication
- Database Security Rules
- Data Validation
- Limited Access

Establishing these security measures will help ensure that users' personal information does not end up in the hands of undesirables.

Software Engineering Paradigm Applied

The model of software engineering which will be used in developing the Deal Theka mobile application will be the Waterfall Model which is one of the most common and traditional models of software engineering. The Waterfall model is sequential, linear and structured and one step of Software Development Life Cycle (SDLC) is achieved before the next step. This will give one a clear picture of what is required, proper documentations and a logical project development.

Waterfall model will be most appropriate to apply in this project because the requirements of the system are also clear, constant and can be ascertained at the initial phase of the project. Since the project is planned as a component of an academic curriculum, the timeline of which is already determined (12 weeks), a more systematic approach, like Waterfall, helps maintain the discipline, record the progress, and complete all the stages within the timeframe. The modularity of the Deal Theka system, such as user management system, service list system, the booking system and the system of control by the administrator, also can be utilized in order to support a strategy of the development in stages.

The making of the Deal Theka application is broken down into various different stages that will lead to the overall success of the system:

1. Requirement Analysis Phase: This is the phase where the requirements of the system are collected and analyzed. The main aim is to gain insight into the issues of the customers and the service providers. The key features are the user registration, the list of service providers, search of the services in the system, booking facilities, and the administration features. This stage will make sure that the system goals are clearly spelt out prior to development.

2. System Design Phase: Once the requirements have been collected after the gathering phase is the system design phase which concentrates on the development of the blueprint of the application. These are system architecture, database structure and data flow diagram, entity-relationship (ER) diagram and the user interface layout designs. Design phase makes sure that all the elements of the system are well-organized and aligned to the project goals.

3. Implementation (Development) Phase: It is during this stage where the application is actually developed. The Deal Theka mobile app is created on Flutter and Dart programming language. They are combined with backend services, like Firebase, to provide user authentication, storage and real-time functions. During this phase, all modules such as user management, service provider module, booking system and the functionality to administer are done.

4. Testing Phase: The system is subjected to rigorous testing after the development process in order to identify and rectify any mistakes or bugs. Various types of testing are unit testing, integration testing and system testing, which are performed to ensure that the application is functional. The stage is crucial in improving the reliability, performance and user experience of the system.

5. Deployment Phase: In the deployment phase, the app will be deployed and tested on Android devices which will be used and shown in the real world. This system is delivered to users and few test run to ensure that the system is working properly in a realistic environment.

6. Documentation Phase: This is the final phase that involves the development of detailed documentation of the entire project like the system design, details of the implementation, user manuals and project reports. Proper documentation will also ensure that the project can be understood, upheld and enhanced in the future.

There are several advantages that can be used in this project and they include; the Waterfall model is well structured, it is easy to manage, milestones are clear and the model is well documented. It enables the developer to concentrate on a phase at a given time making it less complex and better output. Furthermore, as the project requirements are not likely to vary very often, the weaknesses of the Waterfall model, including a lack of flexibility, do not have a profound effect on the development process.

In conclusion, the Waterfall model is a suitable and effective software engineering model when used in the **Deal Theka** project because it enhances order of development, clarity in each phase and a successful transition of the project to be executed within the expected period of time.

Data Models & Diagrams

1. Data Flow Design (DFD)

1.1 Level 0

What is Level 0 DFD?

Level 0 DFD or Context Diagram shows the entire system as one process with interactions between the process and the rest of the world. Deal Theka System is represented as a central process, while all connections with the outer world take place via data flows.

Components of the Diagram

1. Central Process: Deal Theka System

Central element in our context diagram is: Process 0: Deal Theka System. The process includes the system which performs:

- Customer login Service
- listing Searching & filtering
- Service booking
- Services/user management

The process acts as an intermediary between the customer, service provider, and administrator.

2. External Entities

Customer: Customer is the person who:

- Will search for a service
- Will book the service
- See its details

Service Provider: Service provider provides services like:

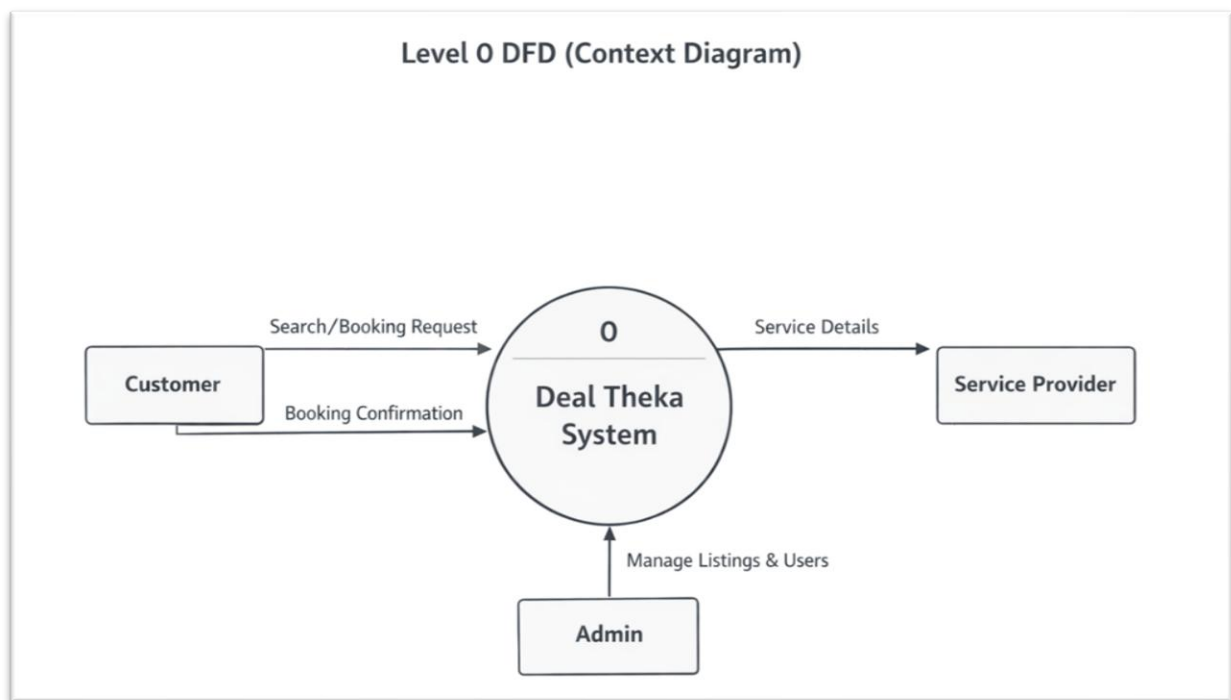
- Electrician services
- Plumbing services
- Driving services
- Carpenter services

They provide the system with: Service details & Availability.

Admin: Admin manages the system and performs actions like:

- Management of the system
- Users/services management
- Activities monitoring

Data Flow



1. Customer → Deal Theka System

Search/Booking Request

- Customer requests:
- Finding services Filtering by category/price
- Booking

Example: Finding electrician services near the customer Booking plumber services tomorrow

2. Deal Theka System → Customer

Booking Confirmation: After the system analyses customer request, it sends the information about:

- Booking confirmation
- Service details

- Contact information of the service provider

Example: Your booking is successful Showing the service details

3. Service Provider → Deal Theka System

Service Details: Providers send the system information about:

- Offered service
- Price Description of the service
- Availability

Example: Electrician services – ₹500. This information will help the system show service details to customers.

4. Admin → Deal Theka System

Manage Listings/Users: Admin does actions like:

- Uploading/downloading/deleting services
- User management
- System monitoring

Example: Removing a fake user Updating the service category

Working of the System (Step-by-Step Flow)

This step-by-step flow should be included in the report

1. The service provider uploads details of the service into the system.
2. Admin controls and updates the data.
3. Customer searches for required services.
4. The system analyses the data and returns the results to customer.
5. Customer books a service.
6. The system confirms the booking to the customer.

This DFD Services platform multi-user interaction between entities Real-time service discovery Limitations of Level 0 DFD (This section needs to be written for good grades) No internal processes described No database included No workflow analysis High-level only overview of the system

Context Diagram of the **Deal Theka** System highlights connections between the central process/system and the outer world. Key data flows and limitations of level 0 DFD are explained.

1.2 Level 1

In the present Level 1 DFD, process 0 represented in Level 0 DFD is broken down into several processes which describe the functionality of the application. Data flow among other processes, external entities, and data stores is described in Level 1 DFD. A Level 1 DFD makes it possible to analyze the internal processes of the software application in detail.

Main Processes in Level 1 DFD

Process 1.0 – User Management

This process describes all operations associated with user management.

Functions:

- User Registration and Login
- User Profile Management
- Storage of the User Data into D1 (User database)
- Sending Information on Services to Other Processes

Data Flow:

- Gets Users Data from the Customer
- Sends Data to D1
- Shares Information on User to Service Search, Booking Management

Process 2.0 – Service Search

The purpose of this process is to enable the search for service.

Functions:

- Receive Search Request from Customer
- Search for Services
- Send the Search Results Back to the Customer

Data Flow:

- Gets search Request from Customer
- Interact with User
- Management to Validate the User Sends Search Results to Booking Management
- Sends Results to Customer

Process 3.0 – Booking Management

This process is the key process in the application and therefore, it is used two times in the diagram.

Functions:

- Receive booking requests from the Customer
- Send Booking Requests to Service Providers
- Receive booking Responses from Service Provider
- Store booking Records

Data Flow:

- Receive Booking Request from the Service Search
- Send Booking Details to Service Provider
- Receive Booking Response from the Service Provider
- Store booking information in D3 (Booking Record)
- Send booking Confirmation to the Customer
- Send booking Record to Admin

Data Stores

D1 – User Database:

- Stores information on Users' Name, Email, Login Credentials
- Used for Validation and Profile Management

D2 – Service Database:

- Stores all the data on the Services
- Used during Listing and Searching for Services

D3 – Booking Records:

- All Booking Transactions
- Used for Tracking Booking History

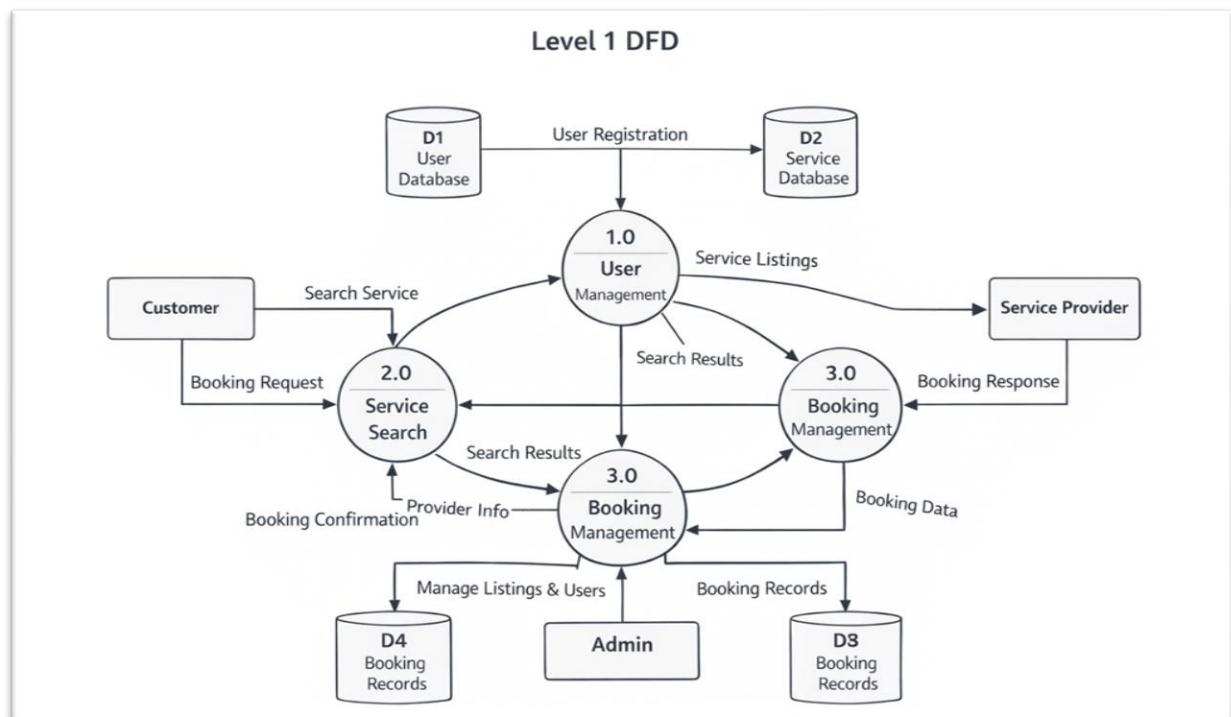
D4 – Booking Records:

- Store Booking Confirmations
- Helps maintain Redundancy and Extended History

External Entities

- **Customer** Search Request and Booking Request Gets Search Result and Booking Confirmation
- **Service Provider** Gets Booking Request Responds with Booking Accept/Reject
- **Admin Managing** the Users and Services Accessing Booking Records

Data Flow Explanation



Explanation of the operation of the system is described below:

Step 1: Customer Sends a Search Request to the Service Search.

Step 2: Service Search gets Validated by the User Management Process and Gets Data Relevant to the User.

Step 3: Generates and Shows Search Results to the Customer.

Step 4: Customer Sends a Booking Request

Step 5: Booking Process gets Request and Sends Booking Details to Service Provider.

Step 6: Service Provider Responds with Accept/Reject

Step 7: Booking Manager Confirms and Stores Booking Record in D3 (Booking Record).

Step 8: Booking Confirmation Sent Back to Customer.

Step 9: Admin Can Access All Data about Users, Services and Booking Records.

Key Features of Level 1 DFD

1. Representations of Internal Processes
2. Defines Interaction Between Different Modules
3. Defines Representation of Data Storage
4. Enhances Understanding of the Flow of the Software Application
5. Useful for Design

2. ER Diagram

The Entity Relationship (ER) Diagram represents the logical structure of Deal Theka system's database. It represents entities, their respective attributes, and the relationships between entities.

With the ER diagram, we develop a systematic approach wherein the data is stored, uploaded and linked to each other. This approach ensures consistency and minimizes repetition saving us time and making the system more efficient.

Entities and Their Attributes

User Entity

Users represent an entity containing all users of the system. Users include customers, the administrators, and others.

Attributes:

User_id (Primary Key): Unique user Identification number that is allocated to every user.

Name: Name of the user

email: E-Mail Address (Unique)

phone: Mobile number

password: Encrypted password

Role: Role gives information on whether the user is an admin or a customer. If customer the role value will be 'customer' while if it's an admin then the value would be 'admin'.

Description: This entity represents all the common user information. Depending on the Role attribute, the user could be a customer or the administrator.

ServiceProvider Entity

Service Provider is an entity that represents all those who provide services.

Attributes:

provider_id (Primary Key): Unique identifier

serviceList: List of services provided

ratings: Customer rating

Description: A Service Provider is linked with a User and provides a service like plumbing, electrical, etc.

Booking Entity

This entity acts as an intermediary entity facilitating interaction between customers and providers of service.

Attributes:

Booking_id (Primary Key): Uniquely defines a booking.

Customer_id (Foreign Key): Links to User

provider_id (Foreign Key): Links to ServiceProvider

booking_date: Defines when the booking was made.

Status: defines the status of the booking, this may contain values 'pending' , 'confirmed' , 'finished'.

Description: This entity will store the detail of each booking made.

Payment Entity

This entity stores the payment details.

Attributes:

payment_id (Primary Key): Uniquely defines a payment.

Amount: Stores the amount of payment.

Status: Defines the status of payment, may contain values 'Paid' , 'Pending' , 'Failed'.

Description: This relates to all booking transaction records where payment feature is enabled.

Category Entity

Category is an entity where services are grouped together.

Attributes:

category_id (Primary Key): Primary key identifier

Category_name: name of category which can be applicable to any of the services (i.e. Electrical, plumbing, etc.)

Description: This entity will help in organizing services and this can also be a factor in optimizing the search function.

Relationships Between Entities

User 'creates' Booking

- One to Many Relationships
- A single user may create many bookings.
- A booking is serviced by exactly one user.

Reason: This would also mean a customer makes multiple service bookings.

User becomes ServiceProvider

- One-to-One Relationship
- The user may also serve as the service provider.

Reason: The registered user also serves as the resource provider of service for the rest of the users.

ServiceProvider accepts Booking

- One-to-Many Relation
- This is acceptable to the C/S architecture. The one service provider may accept many booking.
- Each booking process is attended by only one service provider.

Booking "belongs to" Category

- Relationship: Many-to-One
- Many bookings are related to one Category.

Reason: The three different categories of Booking are given below.

ServiceProvider "provides" Category

- Relationship: Many-to-One
- A Provider can provide a range of services belonging to any category.

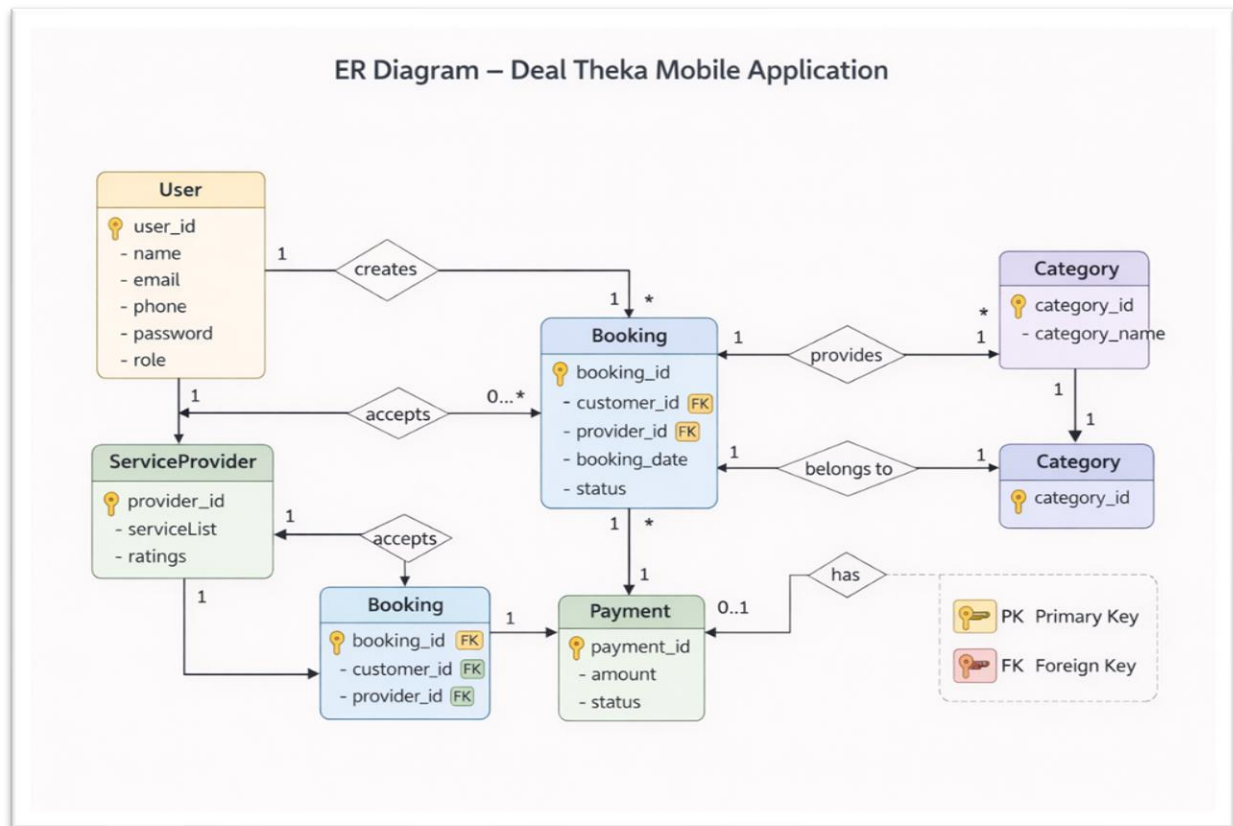
Reason: All the services belong to a specific category and they can provide some service category.

Booking "has" Payment

- Relationship: One-to-One
- A booking may or may not have a payment.

Reason: A payment is optional as the payment is not made when the online payment facility of the system is not functional.

Keys used in the diagram:



Primary key: unique identifying attribute for each of the entities.

Examples:

- user_id
- provider_id
- booking_id
- payment_id
- category_id

Foreign Key: Attributes used to establish links between entities.

Examples:

- customer_id in Booking entity is related to User entity.
- Provider_id in Booking entity is related to ServiceProvider entity.

Cardinality Summary:

- One user: many bookings.
- One provider: many bookings.
- One category: many services/bookings.
- One booking: one provider.
- One booking: zero/one payment.

Process of the ER model (explanation of the flow):

1. A user can register in the system.
2. The registered user may be either a customer or a service provider.
3. The services from service providers are offered under some service category.
4. A customer can then choose their service provider and book for that service.
5. A booking is then made and it is assigned to some service provider.
6. The assigned service provider accepts or rejects the booking.
7. Payment is optionally performed and recorded in the system.

Advantages of the ER model structure:

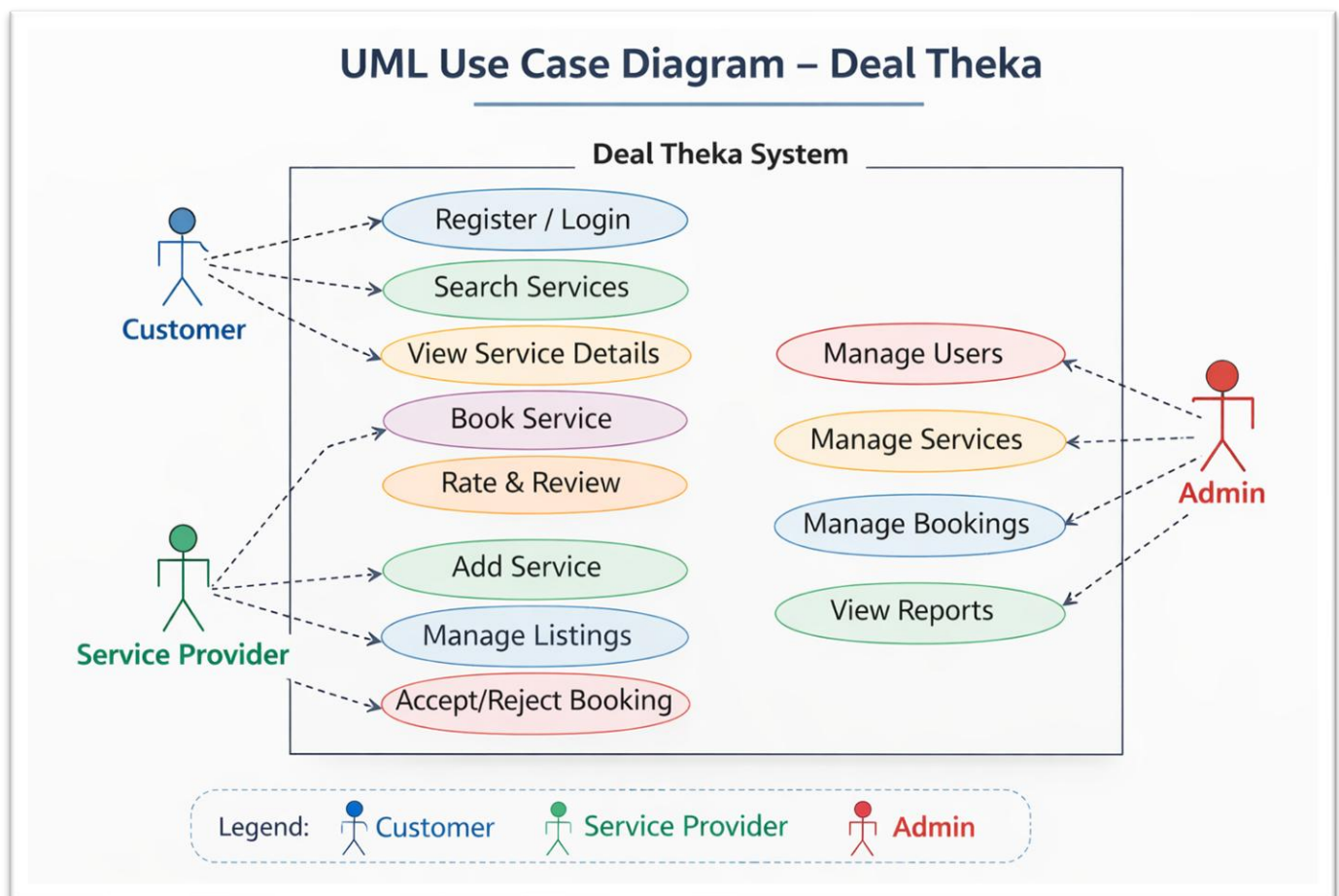
- Separated entities
- Less data redundancy
- Scalable
- Ensures data integrity via primary and foreign keys
- Efficient querying and relationship with entities

Limitations:

- Doesn't explain the implementation.
- Some relations would require normalization.
- Additional Entity 2

The ER Diagram for Deal Theka Mobile Application has properly indicated that the system is designed very clearly. This model explains some primary objects and their relation that are the User, Service Provider, Booking, Payment and category which helped in optimizing data process to make the system better and more efficient.

3. UML Design



In UML, use case diagram depicts the functionality of the system based on user requirements. The use case diagram illustrates the interaction between various types of users (or actors) and the functionality of the system that the user can perform.

As an example, in the context of the **Deal Theka** system, the use case diagram illustrates the interaction between three significant actors, namely, Customer, Service Provider, and Admin.

Component of the use case diagram

System Boundray

The Use Case Diagram has distinct elements. The system is represented by a rectangle labelled "**Deal Theka System.**" Within the rectangle represents everything that is a part of the system, including all use cases.

The purpose of the system boundary is to distinguish:

- Actions performed within the system,
- Actors performing actions outside the system who are interacting with the system.

Roles of the System

1. Customer: The customer is the main user who searches for services and books services.

Responsibilities:

- Customer Authentication
- Search for service offerings
- Read detailed descriptions of service offerings
- Book services
- Provide ratings and reviews for services used

2. Service Provider: The service provider is a user who uses the system to provide a service offering.

Responsibilities:

- Service provider are to add new service offerings
- Manage service listing
- Accept/reject a booking request

3. Admin: The admin is responsible for managing and controlling the system.

Responsibilities:

- Admin can manage users
- Admin can manage service offerings
- Admin can manage service bookings, and view reports

Use Cases and Descriptions

1. Create account or login

Actors: Customers

Synopsis: This use case explains how a user registers an account and logs into the applications securely.

2. Search for services

Actors: Customers

Synopsis: The primary way customers can search for a service is by inputting keywords, categorys, or locations.

3. View service details

Actors: Customers

Synopsis: Customers can view more detailed information about services after conducting a search for them i.e., pricing, descriptions and service providers may also be rated according to their performance.

4. Book service

Actors: Customers

Synopsis: Customers book or redeem services based on their personal requirements. At this point, the customer and service provider have engaged in an interaction.

5. Ratings/reviews

Actors: Customers

Synopsis: After receiving a service, customers can provide ratings, reviews, and feedback on the quality of the services they received. This information is used to improve the overall quality of the services provided.

6. Add Services

Actors: Service Provider

Synopsis: Service providers may add new services to the system by inputting specific service-related data into existing records (category, price, description, etc.) for the continued enhancement of existing records for related service providers.

7. Manage service listing

Actors: Service provider

Synopsis: Service providers are privy to the ability to edit or delete service listings created by them anytime.

8. Accept / Reject Booking

Actors: Service provider

Synopsis: A service provider receives a request for a booking and may either accept or reject the request depending on their availability.

9. Manage Users

Actors: Admin

Synopsis: The admin can view, modify and/or delete user accounts as needed to help maintain the integrity of the system.

10. Manage Services

Actors: Admin

Synopsis: The admin monitors and manages all service listing in order to ensure that they meet an acceptable level of quality and to prevent the misuse of the system.

11. Manage Bookings

Actors: Admin

Synopsis: The admin oversees the tracking and management of all booking activity within the system.

12. View Reports

Actors: Admin

Synopsis: The admin can produce and retrieve reports of user activity, service activity and/or booking activity for analysis and decision making.

Relationships in the Diagram

- Associations connect each actor to specific use cases within the diagram.
- The diagram illustrates the functionalities available to each actor based on their respective usages in the system.
- There are no explicit include or extend relationships illustrated in the diagram.
- However, logically, the following is true:
 - Booking usage is dependent upon login
 - Submittal of booking and rating usage is dependent upon completion of booking.

How the System Works (Use Case Flow)

1. The customer registers and logs into the system.
2. The customer searches for a service and views the service details.
3. The customer reserves the service.
4. Once the reservation request is received by the service provider, the service provider has the option to either accept or decline the reservation.
5. Upon completion of the service, the customer rates and reviews the service.
6. The administrator monitors all users, services, and reservations, and has the capability to produce reports.

Key Features

- The Roles of Users Are Clearly Defined and Unique
- The Software Is Designed with the User in Mind
- The System Supports Service Discovery and Reservation
- Administrators Control and Manage the System

The UML Use Case Diagram of the **Deal Theka** System provides an adequate visual representation of the interactions that occur between the different users and the system. The diagram illustrates the main functionalities such as service discovery, service reservation, service-provider management, and reporting, which were used to develop the system requirements and have laid the groundwork for additional design and development.

SYSTEM DESIGN

The system design stage determines the internal organization and flow of operations of the Deal Theka platform. It translates functional requirements to a well-structured architecture made of modules, data structures, database schema, and user interface elements. This design is aimed at making the system be scalable, maintainable, secure and efficient in terms of dealing with user interactions and processing data. The design is a modular and layered design in which each component performs a particular functionality. This division of concerns enhances the readability of the code, eases debugging, and facilitates future upgrades without impacting on the whole system.

1. Modularization Detail

The Deal Theka system is subdivided into modules, which can be logically independent. The modules each do a certain task and communicate with the other modules via well-defined interfaces.

1.1 User Module

This module deals with the entire system user operations. It also gives the user registration, log-in, profile update, and account management functionality. It makes sure that certain features are not accessed by unauthorized users.

Key Functionalities:

- Registration and login of users
- Profile Management
- Watching and searching services

1.2 Admin Module

The control of the whole system is offered in the admin module.

Key Functionalities:

- Manage users
- Add, update, delete services
- Monitor system activities

1.3 Service Provider Module

Service provider module offers the functionality of addition of new services, updating of existing services and deletion of any existing services.

Key Functionalities:

- Add, update, delete multiple services

- Approve/ deny the user service request
- Track all previous activities

1.4 Authentication and Security Module

This module will provide safe access to the system. It checks the identity of the user and secures confidential information.

Key functionalities:

- User authentication (login/logout)
- Password encryption
- Session management

1.5 Search & Filter Module

This module improves usability, as users can easily locate the appropriate deals.

Key functionalities:

- Keywords search service
- Apply filters (price, category, etc.)

1.6 Notification Module

This module improves user engagement by providing updates.

Key functionalities:

- Alert for new services
- Notifications for updates

2. Data integrity and constraints

The accuracy and consistency of data is important to the dependability of the system. There are several integrity constraints implemented at both the database and application level.

2.1 Entity Integrity

Every table within the database has a primary key which uniquely identifies every record. This will make sure that no duplicates are created and every entity is unique.

2.2 Referential Integrity

Tables are related with each other by means of foreign keys. This will make sure that associated data is consistent throughout the database.

Example:

There is no deal, which does not have a valid user. Thus, the user in the deals table should be identical to a user in the users table.

2.3 Domain Integrity

All the attributes within the database have a designated type and scope of valid values.

Examples:

- Price should be a non-negative number.
- Email should be in a valid format.
- Status fields should have predefined values.

2.4 Not Null Constraints

Some of the areas are obligatory and cannot be vacant.

Examples:

- Username
- Password
- Deal title

2.5 Unique Constraints

There are properties that have to be distinct within the system.

Examples:

- Email ID
- Username

2.6 Application-Level Validation

Besides constraining the database, validation is done at the frontend and back-end to avoid an invalid or malicious input.

3. Database Design

Database is designed using the relational model in order to ensure that data is stored in a structured format, and it is possible to establish relationships between entities.

3.1 Overview of Database Structure

This system involves several interconnected tables such as Users, Deals, Categories and Orders. The database also follows the normalization process in order to reduce redundancy and increase efficiency.

3.2 Table Descriptions

Users Table

Stores information about registered users and service providers.

- user_id (Primary Key)
- name
- email (Unique)
- password (Encrypted)
- role

Service Table

Contains all deal-related information.

- service_id (Primary Key)
- title
- description
- price
- category_id (Foreign Key)
- user_id (Foreign Key)

Categories Table

Used to classify deals into different groups.

- category_id (Primary Key)
- category_name
- category_type

Booking Table

Stores transaction or booking details.

- booking_id (Primary Key)
- user_id
- service_provider_id
- status

3.3 Relationships between tables

- One service provider can generate several services
- There might be any number of services within each category
- A single user can generate several booking requests
- The order is associated with a specific service

3.4 Normalization

This database conforms to the rules of normalization (up to 3NF) in order to:

- Reduce data redundancy
- Increase storage efficiency
- Increase data consistency

3.5 Database Technology

This system makes use of NoSQL database technology such as MongoDB or Firebase to store and query the database.

4. User Interface Design

Deal Theka system boasts an effective, intuitive and efficient user interface. The interface design is targeted at simplicity and ease of use in order to make it easy for users with little technical expertise to use the system.

4.1 Design Objectives

- Create an easy-to-use design
- Ensure that features are easily accessible
- Make it look like other websites on the internet
- Support a number of devices (Responsive Design)

4.2 Design Principles

UI design can be flexible with a wide range of usability principles:

- **Consistency:** Styles, colors and other elements
- **Understanding:** Code texts and instructions
- **Feedback:** System responses to user actions
- **Accessibility:** Communication with all types of users
- **Productivity:** Fewest steps to take to complete tasks

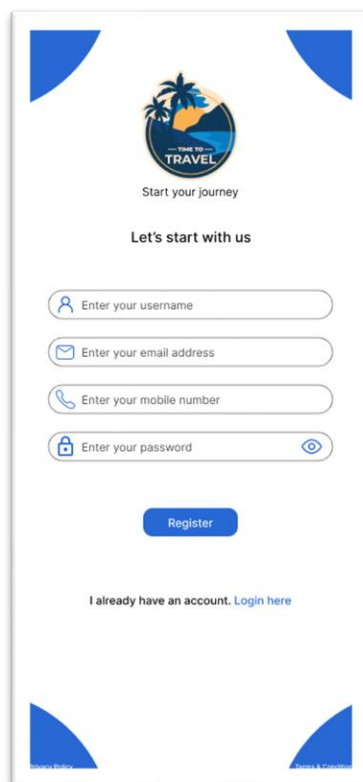
4.3 Key Interface Screens

Login and Registration Interface

Users can access the **Deal Theka** mobile app by logging in or registering via the Login / Registration interface. Effectively providing secure access to the system while offering an easy-to-use and intuitive experience is essential. This interface will allow new users to create their accounts and allow returning users access based on their credentials. The interface has been designed with usability, accessibility, and security in mind to ensure that it provides a positive user experience for all types of users.

Registration Page

The Registration Page is specifically developed for new users who would like to create an account with the service. The Registration Page consists of a very clean and easy-to-understand layout; there are clearly defined (labeled) input fields.

The image shows a mobile app registration screen. At the top, there is a circular logo with a palm tree and the text 'DEAL THEKA TRAVEL'. Below the logo, it says 'Start your journey' and 'Let's start with us'. The form contains four input fields: 'Enter your username' with a person icon, 'Enter your email address' with an envelope icon, 'Enter your mobile number' with a phone icon, and 'Enter your password' with a lock icon and a toggle eye icon. A blue 'Register' button is positioned below the fields. At the bottom, there is a link that says 'I already have an account. Login here'.

Input Fields: The Registration Form Allows For Entry Of The Following Fields:

- **Username:** The user enters his or her full name or preferred username.
- **Email Address:** The email address of the user is used for correspondence and the identification of the user's account.
- **Mobile Number:** A user's mobile number is used as both an identifier for the user and may be used to log-in to or verify against their account.
- **Password:** Passwords are used to protect the user's account from others gaining unauthorized access, and to verify that the individual who signed into the user's account is the authenticated user.

Each field has been purposely designed with placeholder text that will provide a guide to prevent errors while users are entering information. The system will take the following three actions when the user clicks on the prominent Register button at the bottom of the form:

- Validate that all input fields are completed and that no required field is left blank.
- Verify that the email address entered is valid and that all passwords entered into the system meet the minimum password strength requirements.
- Store all User Information securely in the database of the System.

There will also be a link at the bottom of the page that states "I already have an account. Log in here." This will enable users who have an Existing Account to move smoothly from the Register Page back to the Login Page and eliminate any confusion about where to go from that point. The following design features will help users understand how to utilize their accounts:

- Simple, clean design of each form will assist in user-friendly use of the system.
- Adequate spacing between each input field for an overall user-friendly experience
- An Icon will be displayed in each input field to provide a visual reference.
- Consistently using the same color scheme throughout the System's User Interface will maintain the System's overall professional appearance.

3. Login Page

All users of Deal Theka can easily login to their accounts via the login page.

Input Fields: There are two input fields in the login form.

- **Mobile number:** This is the primary identification method for who the user is logging in to.
- **Password:** This is used to log into the user's account.

When you enter your username and password into the input fields (listed above), you will be provided access to your account from the database if your entry was successful by using the login button. Also, when you make an attempt to log in as mentioned above, the database will perform all of the following checks:

1. If the credentials matched what the system has stored, then the user will have access to the user's dashboard.
2. If the credentials do not match what the system has stored, then the user will receive a response of invalid login.

Forgot Password: There is a “forgot password” link that will provide users that have forgotten their password with additional resources to regain access to their account with minimum effort.

Functionalities for Users:

There are two main steps to registering and logging into the system:

Step 1 - First, the user opens the application and goes directly to the registration page.

Step 2 – Next, the user enters their information, submits their registration.

Step 3 - After entered, an application will validate the user information, and is retained in a secure manner.

Step 4 – Users now have access to their account by logging in with their Mobile Number and Password.

Step 5 – Once both credentials are validated, users can access their accounts.

Step 6 – If validation fails, the application will display a message.

Step 7 – Users can also request a password reset.

Security Features:

Your data will be protected as a result of:

- Encryption of password prior to being stored in the database.
- **Data Validation:** The application will check all data entered is valid, and/or free of malware.
- **Authentication:** Your Username and Password are authenticated simply by entering them.
- Only authorized users have access to user accounts.

Usability Features

- Site is user-friendly / easy to use.
- **User-Friendly:** Includes instructions and place holder text.
- **Responsive Design:** All sizes of devices will properly render the application.
- **Navigation:** Users can easily navigate from the Login page and/or Registration Page.

Advantages:

- The application is easy to learn and use for new users.
- Users will have less chance of making an input error due to the guidance of data entry.
- Secure access to the system and satisfy users overall.

The Login and Registration interface of the Deal Theka application has been designed in such a way as to ensure that the system provides a secure, effective, and friendly login mechanism to the users. The interface is fundamental to user interaction, as well as the functioning of the application as a whole.

Dashboard Interface

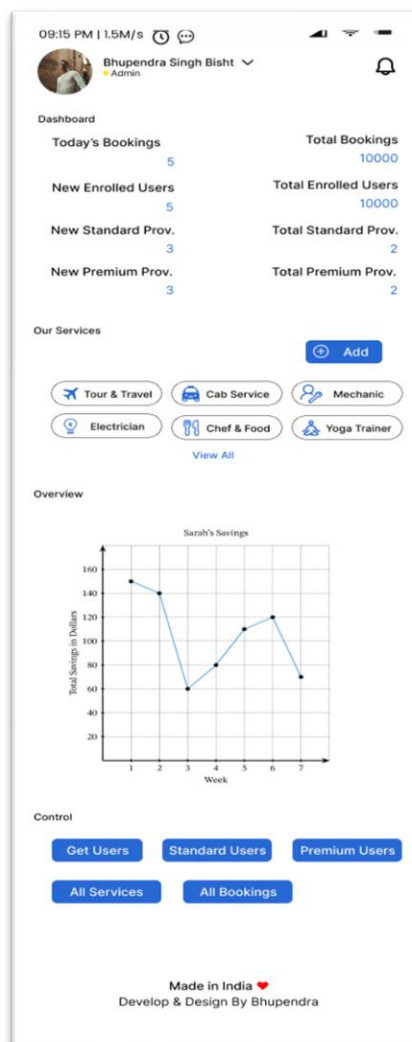
One of the key elements included into the architecture of the Deal Theka application is the Admin Dashboard. It serves as a panel via which the administration can control and monitor all processes and activities that take place in the system.

Dashboard Overview Screen

The screen contains information about the most important aspects and actions performed within the system.

1. User Profile Section

The section contains an administrator's profile including such details as the name and the user role.



2. Statistics Panels

The following panels with statistic data about the system's functioning have been provided:

- **Bookings Today:** the number of bookings performed for today;
- **Bookings Total:** the number of bookings performed in the system in total;
- **Newly Enrolled Users:** the list of the newly created user accounts;
- **Enrolled Users Total:** the number of users created in the system in total;

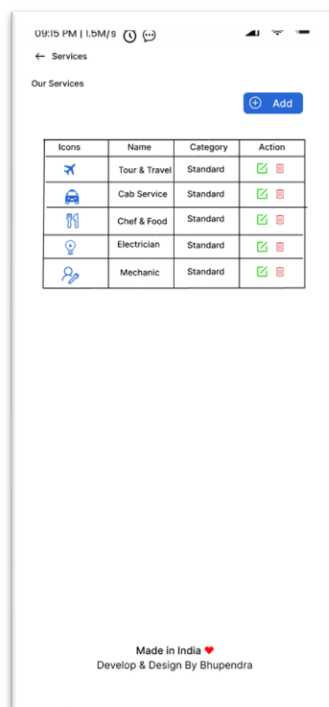
These statistics help conduct a rapid evaluation of activities in the system.

3. Services Overview Section

Here, you can notice different sections that consist of services according to their category, i.e., Tour & Travel, Cab Service, Mechanic, Electrician, Chef & Food, and Yoga Trainer.

- Each service will have a respective tag or button and will provide you ease to access them without any problem.
- Being an admin, you can access all these services or add new services through 'Add' button.

It will become very easy for you to move from one service to another.

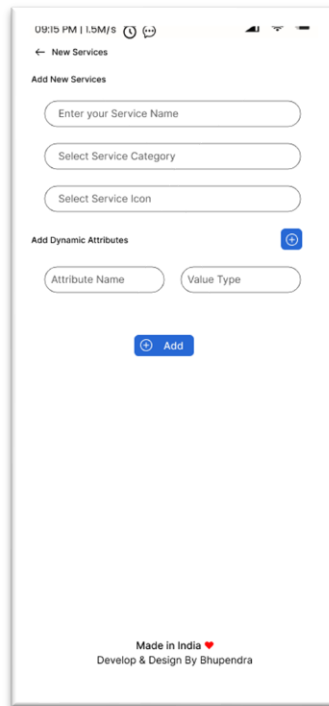


4. Graphical Analysis

Further, there will also be an overview graph in the dashboard through which you can see the graphical analysis in a specific period such as weekly.

- Graphical analysis will include all the data within a certain time period (weekly).
- Through graphical analysis, you can conduct analysis properly.

Graphical analysis is indeed very important for making decisions.



The screenshot shows a mobile application interface for adding new services. At the top, the status bar displays '09:15 PM | LSM/S' and a battery icon. Below the status bar, there is a back arrow and the text 'New Services'. The main section is titled 'Add New Services' and contains three input fields: 'Enter your Service Name', 'Select Service Category', and 'Select Service Icon'. Below these fields is a section titled 'Add Dynamic Attributes' with a blue plus icon. This section contains two input fields: 'Attribute Name' and 'Value Type'. At the bottom of the form is a blue button with a plus icon and the text 'Add'. At the very bottom of the screen, there is a footer that reads 'Made in India' with a red heart icon, followed by 'Develop & Design By Bhupendra'.

5. Control Panel

Following points will make your administration process quite simple and convenient in the dashboard:

- **Get Users:** Seeing all users
- **Standard Users:** Standard users filtering
- **Premium Users:** Premium users management
- **All Services:** All services in one go
- **All Bookings:** View and manage all bookings

Functional Workflow

- Functional Workflow of the Administration Dashboard is shown below:
- Administrator logs in into the system
- System overview statistics will be shown on the dashboard
- Administrator manages booking, users and services
- Administrator navigates to service management where he sees and manages the services
- Administrator creates new services using the form
- New record appears in the database

Design & Usability Features

- System with neat design which is easy to use
- Adapted to work on different devices
- Sections and navigation are well structured
- Graphical interactions like the use of icons
- Quick action buttons present

Advantages of Admin Dashboard

- System management tool in one place
- Makes user and service management easy
- Easy monitoring of system performance
- Facilitates data-based decision making
- Dynamic attribute services management

The Administration Dashboard in the **Deal Theka** application is a very useful feature that is vital in system management.

Service Listing Platform

Service Listing Platform constitutes an integral part of the **Deal Theka** mobile app. This module helps the service provider perform service-related operations and enables tracking of booking and payments by the system. Service Listing Platform serves as the work level for managing services through the execution of booking and payment operations. This module will facilitate smooth operation by both the parties along with proper data handling and management.

Add New Service Interface

This interface helps users add a new service listing into the system by filling out a form. This interface is designed in a flexible and easy-to-use manner.

09:15 PM | 1.5M/s

← New Services

Add New Services

Enter your Service Name

Select Service Category

Select Service Icon

Add Dynamic Attributes

Attribute Name

Value Type

Add

Made in India ❤️

Develop & Design By Bhupendra

1. Input Fields

The fields available in this interface include the following:

- **Service Name:** Entering the name of the service.
- **Service Category:** Selecting the category under which the service falls (cab service, electrician services, food services, etc.).
- **Service Icon:** Selecting an icon to represent the service.

2. Dynamic Attributes Feature

One notable feature in the above interface involves adding dynamic attributes to the service:

- **Attribute Name:** The field in which we input the characteristic of the service (seats, duration, tools required, etc.).

- **Value Type:** The type of value inputted (text, number, etc.).

This feature will help the system accommodate different services with different characteristics without any changes to the database.

Service Completion Interface

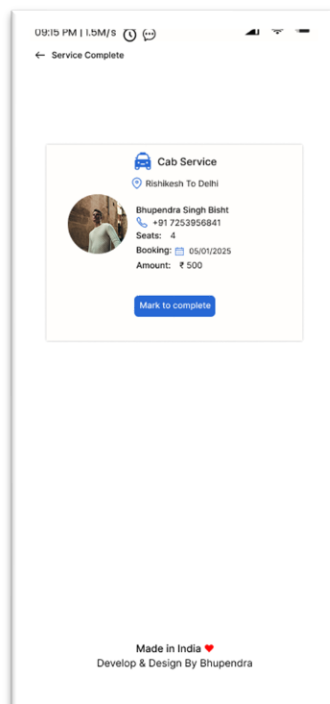
The service completion interface is available to all service providers and is utilized for tracking their current active bookings and marking them as complete.

1. Display of booking information

The information displayed on this interface shows the details of the current ongoing service:

- Name of Service (Cab Service)
- Route of Service or Service Description (Rishikesh to New Delhi)
- Customer Information (name and telephone number)
- Date of Service Booking
- Service Price / Cost

This detailed information assists the service provider to confirm the booking prior to marking the service as complete.



2. Function of marking as complete

To mark the service, complete a "Mark as Complete" button has been incorporated into the service completion interface. Upon clicking the "Mark as Complete":

- Booking Status will be updated in the system.
- A record of the service being marked as complete will be established in the system.
- The system may potentially initiate the processing of payment or update reporting metrics related to that booking.

This feature of marking services as complete will allow for accurate tracking of completed services.

Payment History Interface

The Payment History interface will present the service provider with an overall view of their income and transactions.

1. Dashboard Summary

At the top of the Payment History interface, there are Key Performance Indicators (KPIs) which will provide an immediate overview to the Provider regarding their current performance.

These KPIs will include:

- Daily Earnings - Total Earnings for the Day
- Weekly Earnings - Total Earnings for the Week
- Total Earnings – Total Overall Earnings
- Active Jobs - Number of Jobs Accepted & Out for Service
- Completed Jobs - Number of Jobs Completed
- Cancelled Jobs - Number of Jobs Cancelled

The above KPIs provide a concise and straightforward method for the provider to quickly identify how they are doing

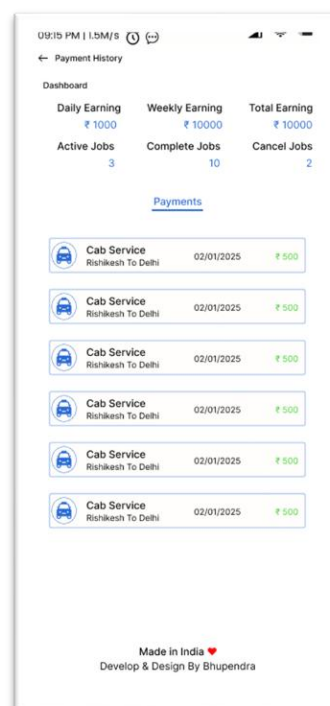
Payment Record List

Just below the above KPIs the Payment History interface will provide the provider with a detailed list of each of their individual transactions, such as:

- Service Type
- Service Route or Description
- Date of Service
- Payment Amount

This will provide the provider with a detailed record of all of their income and transactions.

Functional Workflow



The Payment History interface is part of the overall workflow of using the Service Listing Platform. This includes:

1. Adding a new service by using the Add Service interface.
2. When the service has been added, it will become visible to customers in the system.
3. When customers want to use that service, they will book it.
4. The service provider will view the customer's booking in the Service Completion interface.
5. After the service has been provided, the service provider will complete the service in the Service Completion interface.
6. The Service Listing Platform will change the status of the customer's booking accordingly and the Service Listing Platform will write a transaction for the service provider.
7. The Payment History section will include the transaction information for the service provider.

The Deal Theka Application has its Service Listing Platform that is an all-in-one module for complete management of all services from the original point of creation through to completion and the tracking of payments during that process. The Service Listing Platform will offer service providers with such flexibility, efficiency and transparency that they will be able to manage services effectively and provide customers with a seamless experience. Furthermore, Service Listing Platform encompasses an extremely vital component for the success and growth of the overall system.

4.4 Technologies Used

The application makes use of the following technologies:

- App was developed using Flutter
- The backend was implemented using Firebase

4.5 Responsive Design

The interface can be optimized to several screen sizes in order to cater for all different screen sizes.

4.6 Navigation Flow

The flow of navigation within the system follows the following pattern:

User → Login → Dashboard → Browse Services → View Details → Perform Action

The architecture of the implementation of the system of **Deal Theka** is efficiently constructed. This system can easily be scaled and managed since it involves the disaggregation of the system into modules, ensuring data integrity, normalization of the database and intuitive interface.

Testing of Project

Testing is a critical phase in the software development lifecycle that ensures the correctness, reliability, and performance of the system. The Deal Theka platform was tested using a combination of structured testing techniques and strategies to identify defects, verify functionality, and validate system behavior under different conditions. The testing process was carried out at multiple levels, including unit testing and system testing, to ensure that both individual components and the complete system operate as expected.

1. Testing techniques and Testing strategies used

To make the testing process effective and comprehensive, both manual and functional testing was done.

1.1 Black Box Testing

Black box testing aims to test the functionality of a system regardless of its internal code.

Application in Deal Theka:

- Testing login with correct/incorrect credentials
- Test deal search and filtering functionalities
- Validate form entries such as empty fields, incorrect email address

Advantages:

- Guarantees that the software meets user requirements
- Helps to identify defects and gaps in the system

1.2 White Box Testing

White box testing is done by testing the internal workings of the program, including logic and code structure.

Application in Deal Theka:

- Conditional testing in the user authentication
- Validation of database queries and API calls
- Test control flow within individual modules

1.3 Unit Testing

Unit tests involve testing an individual component in isolation from other modules.

Modules tested:

- User module (for authentication)
- Deal modules for creating/deleting/dealing deals

- Search/Filtering functionality of the deals

1.4 Integration Testing

This type of testing aims to ensure the proper interaction between modules

Examples:

- Interaction between User module and database
- Validating performance while running normally
- Consistency and accuracy of data between modules

1.5 System Testing

System testing focuses on checking the behavior of the entire system.

Focus areas:

- Finding errors while performing end-to-end operations
- Performance under normal usage
- Data consistency across modules

1.6 Validation and Verification

- **Verification:** Ensures that the system has been implemented properly (according to the design)
- **Validation:** Ensures that the developed system meets users' requirements

2. Testing Strategy

Testing strategy adopted is a bottom-up approach where testing starts from individual modules and moving upwards towards integration and testing the system as a whole.

Steps Followed:

1. Testing of Individual Modules
2. Integration Testing among various modules
3. System testing the complete system
4. Testing for basic functionality and performance

3. Test Reports for Unit Test Cases

Unit tests were conducted for some modules to test the behavior of individual functions in the system.

3.1 Unit Test Case Table

Test Case ID	Module	Input	Expected Output	Actual Output	Status
UT01	Login	Valid email & password	Login successful	Login successful	Pass
UT02	Login	Invalid password	Error message	Error message	Pass
UT03	Registration	Empty fields	Validation error	Validation error	Pass
UT04	Deal Creation	Valid data	Deal added	Deal added	Pass
UT05	Deal Creation	Missing title	Error message	Error message	Pass
UT06	Search	Keyword input	Relevant results	Relevant results	Pass

3.2 Observations

- All core modules performed as expected
- Validation checks successfully prevented invalid data
- No critical errors were found during unit testing

4. Test Reports for System Test Cases

System testing was conducted to evaluate the behavior of the complete application.

4.1 System Test Case Table

Test Case ID	Scenario	Input	Expected Result	Actual Result	Status
ST01	User Login Flow	Valid credentials	Dashboard displayed	Dashboard displayed	Pass
ST02	User Registration	Valid details	Account created	Account created	Pass
ST03	Browse Deals	Click on deals page	Deals list shown	Deals list shown	Pass
ST04	View Deal Details	Select a deal	Details displayed	Details displayed	Pass
ST05	Admin Add Deal	Valid input	Deal added	Deal added	Pass
ST06	Invalid Input Handling	Empty form	Error message	Error message	Pass

4.2 Performance Testing (Basic)

- System response times were satisfactory in normal usage conditions
- System navigational response was fast without any significant delays
- Databases queries worked fine

4.3 Security Testing (Basic)

- Passcode encrypted in system database
- Unauthorized access limited by role
- System verifies user inputs and prevents injections

5. Challenges Faced During Testing

- Testing of edge cases for inputs provided by the users
- Frontend and backend synchronization
- Debugging of API calls

The test run showed that the created system was functional enough and met all criteria and specifications. Most errors and bugs found during testing were fixed. Therefore, system testing proved that the created system can be used successfully.

System Security Measures

Security measures play an important role when creating modern web apps that work with user data. Deal Theka system consists of several security features to keep the users' data private and protected from various attacks. The security measures used by this web app follow the approach called Defense in Depth. It implies implementing several protections at different levels of the system.

The system uses several protection layers on its frontend, backend, and even on the level of database interaction.

1. Authentication and Authorization

1.1 Secure Authentication Mechanism

The system provides a reliable method for verifying the user identity.

- User needs to enter email and password correctly
- Password is never stored in clear text
- Verification uses comparison with encrypted password

1.2 Password Hashing

All passwords are saved as hashes instead of clear text.

- This way the password cannot be restored
- Password administrators also do not see passwords
- The common algorithm used is bcrypt

1.3 Role-Based Access Control (RBAC)

Different roles for admins and regular users are implemented.

- Unauthorized actions are restricted at the backend level
- The administrator has full privileges to create/delete deals and users
- Regular user may use only his personal data

2. Data Security Measures

2.1 Data Encryption

Sensitive data is protected using encryption techniques before storage and transmit the data.

- Sensitive data is encrypted using HTTPS protocol
- Data safety during its transfer is ensured

2.2 Input Data Verification and Sanitization

To prevent malicious input, strict validation is applied at both frontend and backend levels.

- Verification includes the presence of needed fields
- Verification of data formats (email, numeric, etc.)
- Removal of potentially harmful elements

2.3 Protection Against SQL Injection

The system uses parameterized queries and ORM-based queries to prevent SQL injection attacks.

- The user inputs are not inserted into SQL queries
- Database interaction happens safely

3. Session and Token Management

3.1 Session Management

- The system manages user sessions
- Session expires after some period of inactivity
- Protection from hijacked sessions

3.2 API Tokens Usages

Tokens are needed for accessing the API if used.

- Token is necessary for authorization to access specific route
- Token expires after some period of time

4. Protection from Web Attacks

4.1 Prevention of XSS Vulnerability

- Received input is sanitized before rendering
- Bad HTML is blocked

4.2 CSRF Tokens

- Tokens used in forms for protection
- Ensure the request came from a user

4.3 Brute Force Attack Prevention

- Number of attempts is limited
- Temporarily locks user account

5. Database Security

5.1 Access Control

- Database access is restricted by using credentials.
- Back-end is able to access database only
- Any other access attempts are blocked

5.2 Data Backup

- Regular backups are maintained
- Backup helps to restore system in case of data loss

5.3 Data Integrity

- Use Primary and Foreign Keys for integrity
- Avoid inconsistent or orphaned data

6. Error Handling and Logging

6.1 Secure Error Messages

- Error message contains no confidential information
- Generic errors displayed to users

6.2 System Log

- System activity is logged
- Suspected activity can be investigated further

7. Environment Security

- Application hosted at secure platform
- Environment variables contain API keys and DB credentials
- Application runs over HTTPS connection

8. Limitations and Future Improvements

While this web app uses some security measures, there are also future improvements to make:

- Two-factor Authentication (2FA)
- More advanced intrusion detection systems
- Role Hierarchy and permissions for each role
- Security Audit and Penetration testing

The **Deal Theka** system consists of several security features that protect user data. The application uses several security mechanisms including authentication, data encryption, input sanitizing, and protection from various web vulnerabilities.

Cost Estimation of the Project along with Cost Estimation Model

Cost estimation is a significant factor during the project planning phase and assists in estimating the total resources required for developing, deploying, and maintaining the deal theka system. This project is about software development, software hosting, maintenance, and other costs.

Cost Estimation Model Used

For this project, we use the **COCOMO Model (Constructive Cost Model)**.

Why COCOMO?

- Easy and frequently used model in the academic field
- Provides effort estimation by size of the project
- Applicable to small and medium-size projects such as Deal Theka

COCOMO Formula

$$\text{Effort (Person-Months) } E = a \times (\text{KLOC})^b$$

Where:

- **E** = Effort required
- **KLOC** = Thousands of lines of code
- **a, b** = Constants based on project type

Project Type Selection

The **Deal Theka** project is categorized under Organic Mode (small team, less complex system)
Typical constants:

- **a = 2.4**
- **b = 1.05**

Estimation Assumptions

Parameter	Value
Estimated Lines of Code	10,000 LOC
KLOC	10
Team Size	2–3 members
Development Time	3–4 months

Effort Calculation

$$E = 2.4 \times (10)^{1.05}$$

$$E = 2.4 \times 11.22 = 26.93 \text{ Person-Months}$$

Development Cost Estimation

Assuming:

- Cost per person per month = ₹15,000 (estimation)

$$\text{Total Cost} = 26.93 \times 15000$$

$$\text{Total Cost} = ₹4,03,950$$

Hardware & Software Cost

Component	Cost (INR)
Development System (Laptop/PC)	₹50,000
Internet Charges	₹2,000
Hosting (Render / VPS)	₹3,000/year
Domain Name	₹1,000/year
Software Tools (Open Source)	₹0
Total	₹56,000

Maintenance Cost

Type	Cost
Annual Maintenance	₹20,000
Bug Fixing & Updates	₹10,000
Total	₹30,000/year

Total Estimated Cost

Category	Cost
Development Cost	₹4,03,950
Hardware & Setup	₹56,000
Maintenance (1 year)	₹30,000
Total Project Cost	₹4,89,950

As we can see, the cost of the Deal Theka project is nearly ₹4.5 lakh to ₹5 Lakh. As this is an academic project using open-source tools, this cost can be minimized significantly.

Reports

The **Deal Theka** system utilizes reports for accurately structuring and providing actionable data that has been created from all available data in the system. Administrators rely upon these reports as a means to monitor all activities within the system, identify any performance issues within the system and provide the basis for informed decision making.

The reporting system will capture data from all modules of operations including user, service and booking related information and subsequently display the information in a structured and easy to read format. Reporting provides transparency and strengthens control over, as well as increases efficiency for, the operation of the **Deal Theka** System.

The system will provide reports that are dynamically generated from real time data stored in the Database (**Firestore**) and provide timely information as it relates to current state of operations.

Objectives of Reporting System

The reporting module of the system has been created with several objectives:

1. To provide the ability to monitor how well the system performs and its growth.
2. To track how users engage with the system and their activities.
3. To analyze usage and popularity of the services offered.
4. To manage booking transactions in a manner that is both accurate and timely.
5. To assist in making decisions based on data insights.

Types of Reports within the Deal Theka System

The Deal Theka system generates three primary types of reports:

1. User Report
2. Service Report
3. Booking Report

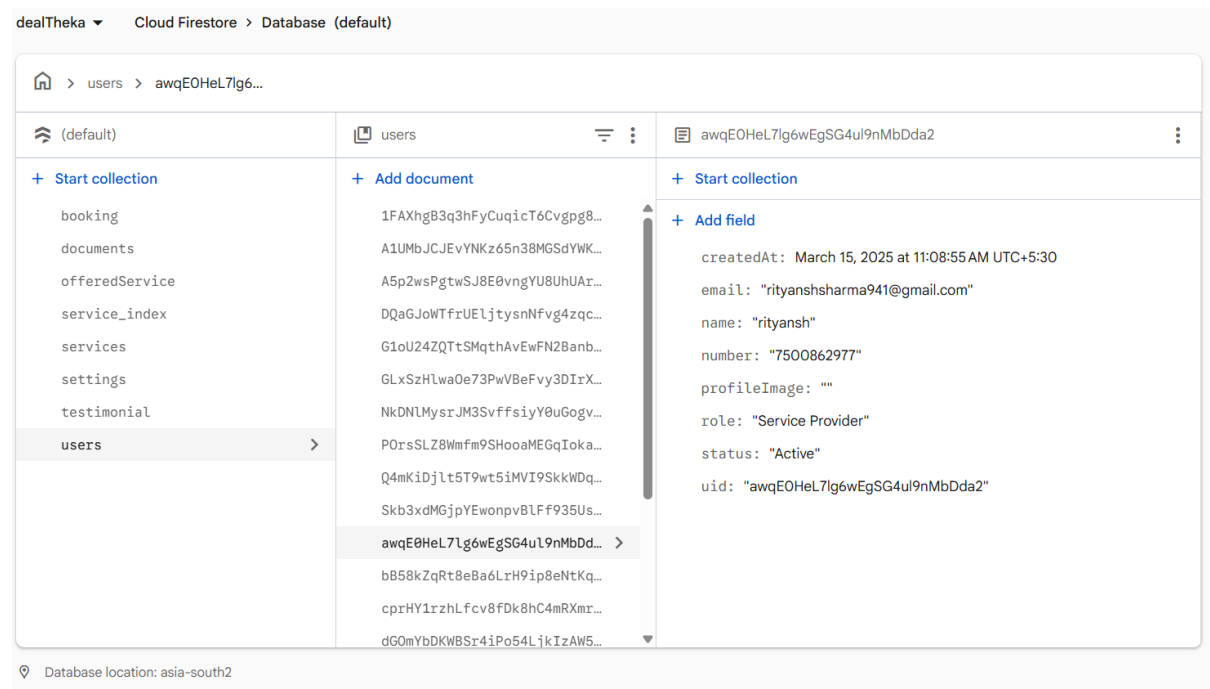
Each report is designed for a specific purpose within the DealTheka system and contains detailed information regarding a number of aspects related to the operation of the DealTheka system.

User Report

Content This report summarizes all the users for the app. This will allow the administrator to track the number of users, user activity, and the roles of users.

Data Source: This report is created from the Users Collection in Firebase Firestore.

Included Fields: The User Report collection contains the following data:



- **User ID:** The unique ID assigned to each user.
- **Name:** The user's full name.
- **Email:** The user's email address.
- **Phone Number:** user's phone number.
- **Role:** User is either a customer or a service provider.
- **Status:** The current status of user account (active/inactive).
- **Date Created:** Date user account was created.

Search by email address, phone number, or user UID Add user				
Identifier	Providers	Created ↓	Signed In	User UID
cc@dv.con		May 7, 2025	May 7, 2025	GLxSzHlwaOe73PwVBefVy3D...
dgs@zc.vbn		May 7, 2025	May 7, 2025	jrQJ3dBcQJfTiv3Kkl1qong9ez...
baba@ydw.ckm		May 6, 2025	May 6, 2025	1FAXhgB3q3hFyCuqicT6Cvvp...
vshhz@gm.jdbd		May 6, 2025	May 6, 2025	Skb3xdMGjpYEwonpvBlFf935...
bhupendrabisht8@gm...		May 5, 2025	May 5, 2025	A1UMbJcJEvYnKz65n38MGS...
ritlu6666@gmail.com		Apr 17, 2025	Apr 17, 2025	POrsSLZ8Wmf9SHooaMEG...
siraj8897@gmail.com		Apr 13, 2025	Apr 13, 2025	qZ09rERlm2VmXi9iX5NLS5G8...
rai6878@gmail.com		Apr 13, 2025	Apr 13, 2025	cprHY1rzhLf8fDk8hC4mRX...
ritu6666@gmail.com		Apr 13, 2025	Apr 13, 2025	Q4mKiDjlt5T9wt5IMVI9SkkW...
priyatiwari5447@gmail....		Mar 27, 2025	Mar 27, 2025	ePOZFwqg8pNlh8JcvB75cWX...
augustesharma077@g...		Mar 26, 2025	Mar 26, 2025	A5p2wsPgtwSJ8EOvngYU8Uh...

Functionality

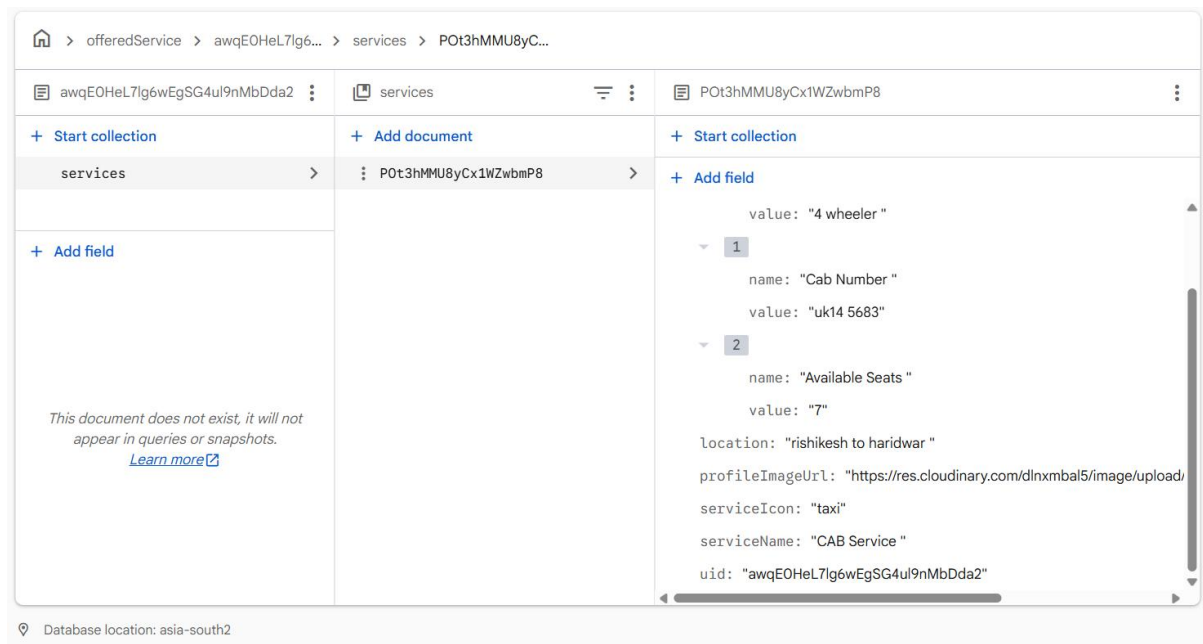
- See all users of the system.
- View users by role (Customer / Service Provider).
- Determine active versus inactive users, and identify trends in user growth.

Purpose

- Manage User Accounts.
- Observe how users are adopting the Platform.
- Track user growth trends.
- Secure/use the System.

Service Report

Service Reports provide insight into the services offered in the system by providing a summary of services provided by the Service Providers to Customers. The report documents/records the types of services, their availability, and the contributions of each service provider with respect to the type of service.



Data Source

The service report is developed using the Services and the Services Provided data collections within the database.

Fields Included

Service ID = Unique ID for the service.

Service Name = The name given to the service.

Categories = The type of service (e.g., taxi, electrician).

Service Provider ID = ID associated with the service provider.

Service Location = The location where the service will be provided.

Attributes = Characteristics of a service (e.g., number of passengers, cost).

Status = Active / inactive.

Functionality

- List All Services Available
- Categorize Services
- Analyze the most commonly used services
- Monitor the availability of services

Purpose:

- To define the service's distribution model.
- To identify services that are experiencing high demand.
- To enhance the service's overall offerings.
- To manage the service listings within the platform.

Booking Report

The Booking Report provides a complete overview of all booking transactions made through the platform. This report is vital for measuring total use of the overall platform, as well as providing a way to measure the financial aspects of the service.

🏠 > booking > OEUrO4Xwj1lBl...		
🏠 (default)	📄 booking	📄 OEUrO4Xwj1lBlVwz9tIH
+ Start collection	+ Add document	+ Start collection
📄 booking > documents offeredService service_index services settings testimonial users	📄 OEUrO4Xwj1lBlVwz9tIH > hpMYUH1MVg8JJZ6cCY0c	+ Add field createdAt: May 11, 2025 at 5:13:18 PM UTC+5:30 date: "2025-05-11" icon: "auto_rickshaw" location: "rishikesh" serviceId: "eHbarV1sX9PFbmP4pV8Q" serviceIndexId: "XTM29rQT2BwHejJcMC5n" serviceName: "AUTO RICKSHAW " serviceProviderId: "jrQJ3dBcQJfTiv3Kkl1qong9ezZ2" status: "Pending" userId: "GLxSzHlwaOe73PwVBeFvy3DlrXo1"

📍 Database location: asia-south2

Source of Data: All data for the Booking Report comes from a Booking Collection in Firebase.

Fields Included: The fields in the Booking Report will include:

- a) **Booking ID:** A unique number assigned to each booking
- b) **User ID:** A unique number assigned to the user who made the booking.

- c) **Service Provider ID:** A unique number assigned to the Service Provider associated with the booking.
- d) **Service Name:** The name of the service being booked.
- e) **Date:** The date of the booking.
- f) **Location:** The service location.
- g) **Status:** Whether the booking is (1) pending, (2) completed or (3) cancelled.
- h) **Payment information:** The amount paid and if the payment has been completed (if applicable).

Functionality: The Booking Report will allow for the following operational functions:

- a) Monitor and track all of the booking activity in the Platform.
- b) Monitor and track due dates of all bookings in the booking system.
- c) Monitor and track each user's daily and weekly activity with their bookings.
- d) Identify which bookings have been completed and which have been cancelled.

Purpose: Purpose of Using A Transaction Management Reporting System

- Transaction Management
- Analyzing Performance
- Reporting Revenues
- Improving The Efficiency Of Services

Report Generation System

The following procedure will be used when creating reports:

- Data will be collected from Firebase collections (User, Service and Booking);
- The data collection from Firebase collections will be done using structured queries;
- Data must then be converted and formatted into a report that makes sense;

Finally, reports can be shown in tables and charts.

Benefits of Using A Reporting System

- Provides Instantaneous View Of What Is Happening
- Better Decisions Can Be Made
- Better Monitoring of The Reporting System
- Better Analysis of The Reports
- Identifying Issues at An Early Stage

Future Enhancements of Reports

The Reporting Module could be made better by adding:

- Graphs & Charts to The Reporting Module (Dashboard);
- Ability To Export Reports to PDF/Excel;
- Analytical Data Prediction with A.I.;
- Predictive Insights; +
- Real-time Alerts.

The Deal Theka System's reporting module is critical to monitoring and managing all users, services, and transactions. It provides structured and meaningful information to enable administrators to make effective decisions and improve the system's overall performance. Real-time database reports add to the effectiveness and scalability of the reporting module as well.

Conclusion

The “Deal Theka” mobile application has been successfully created and released as a connected digital platform for local service providers and users to link up with one another, while also being ordered through a simple-to-use, well-organized, and efficient way. The goal of this project was to link users who need local services to local workers who do not have the opportunity to work on a regular basis. By deploying this new technology, the “Deal Theka” application provides significant evidence regarding the use of modern-day technology to address local community and world issues.

The “Deal Theka” application also addresses a major issue within this area - the absence of consistent, dependable methods for finding local service providers, particularly in semi-urban and rural regions. Users in these regions have traditionally sought to find service providers in their area through informal channels, such as referrals from family/friends, however, these informal channels can often come with issues related to reliability, accountability, and timeliness. On the other hand, many skilled workers that reside in these same regions have great challenges reaching a larger potential target audience because they do not have access to a digital medium or means to effectively market their skills. The “Deal Theka” application successfully solves the various problems mentioned above by providing a central marketplace where all service providers will have their services posted, searchable, and available to book all through the same programme.

Flutter was utilized to develop a front-end while Firebase provided the backend services. This type of structure is able to allow the solution to be scalable, responsive and cross-platform. With a modular design that includes modules for Users, Service Listings and Booking, the solution has good maintainability and provides the opportunity for future enhancements.

Features of the solution include search for services by filter criteria, Booking Management, and Administration Control providing efficiencies for managing operations and improving customer experience. The potential sociological impact of the application is significant. In addition to assisting Service Providers to digitise and ultimately become more visible and provide more employment opportunities, it will also save customers time and effort in searching for or booking a service after conducting a search to identify a suitable (pre-verified) service. Connecting local Service Providers to an online marketplace and providing fast access to verified services for customers will drive economic development at the local level and support the larger digitalisation initiative within developing regions.

Various Software Engineering Principles and Practices were followed while working on developing this Project; such as system analysis, requirement specification, design modeling (DFD, ER diagrams, UML) and testing. Using the Waterfall Model has enabled developing in a Structured/Organized Way. The project has also provided a wealth of learning opportunities

with Mobile App Development/Database Management/UI/UX Design and Cloud-Based Backend Integration.

The current system does have limitations, such as the lack of Integrated Online Payment Systems, Advanced Recommendation Systems, and GPS Based real-time Tracking, which can be further enhanced and/or improved to develop the platform (take it to the next level).

"In conclusion, the Deal Theka Application can successfully meet its purpose by creating an efficient, reliable, and scalable digital marketplace for local services. This demonstrates the Real-World Application of Theory Concepts learned through the MCA Program and further supports the ability of Technology to solve real-world societal challenges." There is a lot of potential for the application to evolve into an incredibly powerful utility to Empower Local Communities and improve accessibility to services as enhancements are made and it is used in the Real World.

Future Scope and Further Enhancement

The **Deal Theka mobile application** provides a platform that connects local service providers with customers in a simple and efficient way. Although the current system focuses on basic service discovery and booking functionality, there are several opportunities for future improvements and expansion.

1. **Online Payment Integration:**

In future versions, the application can integrate secure digital payment systems such as **UPI, credit/debit cards, and digital wallets**. This will allow customers to make payments directly through the application, making the service booking process faster and more convenient.

2. **GPS Based Location Services:**

Location-based services can be added to help users find nearby service providers more accurately. By using **GPS and map integration**, the system can display service providers based on the user's current location.

3. **Rating and Review System:**

A rating and feedback mechanism can be implemented where customers can rate and review service providers based on their experience. This will improve service quality and help users select reliable providers.

4. **Real-Time Chat System:**

A messaging feature can be introduced to allow direct communication between customers and service providers. This will help users clarify service details, pricing, and availability before confirming a booking.

5. **Multi-platform Support:**

Currently, the application focuses on android mobile platforms. In the future, a web-based version, IOS mobile version of the system can be developed so that users can access services from computers and tablets and Apple smart phones as well.

6. **AI-Based Service Recommendation:**

Artificial Intelligence techniques can be used to recommend services to users based on their past searches and booking history. This will improve user experience and make service discovery faster.

Overall, these enhancements can transform Deal Theka into a **large-scale digital marketplace for local services**, improving accessibility, efficiency, and employment opportunities for local workers.